

Project Data Page:





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MP Nitrogen Buffer Vessel(V8602)

DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\MP NITROGEN BUFFER VESSEL-V8602 7%.pvdb

Date of Analysis : Apr 20,2026 1:09pm

PV Elite 28, January 2026



Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1000.000 mm. ID	
Vessel Design Length, Tangent to Tangent	2000.00 mm.
Distance of Bottom Tangent above Grade	10000.00 mm.
Specified Datum Line Distance	0.00 mm.
Internal Design Temperature	82 °C
Internal Design Pressure	30.500 bars
External Design Temperature	0 °C
Maximum Allowable Working Pressure	30.5 bars
Shop Test Pressure	47.678 bars
Field Test Pressure	39.650 bars
Required Minimum Design Metal Temperature	-12.0 °C
Warmest Computed Minimum Design Metal Temperature	-29.0 °C
Wind Design Code	ASCE/SEI 7-16
Earthquake Design Code	ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-105	...	...	K03504	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Leg Baseplate		...	...		
Leg Bolting		...	...		

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
bottom Dish	30.500	0.00	33.700	1.6000	No	No
Shell	30.500	0.00	33.100	1.6000	N/A	No
Top dish	30.500	0.00	33.700	1.6000	No	No

Liquid Level: 2500.00 mm. Dens.: 0.000 kg./cm³ Sp. Gr.: 0.001

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Ellipse	50.0	50.0	20.0	16.0	14.6	...	0.85	0.85
Cylinder	1950.0	1900.0	16.0	16.0	14.9	...	0.85	0.85
Ellipse	2000.0	50.0	20.0	16.0	14.6	...	0.85	0.85

Local Stress Analysis Results:



Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
Lift Lug	WRC-107/537	0.063	n/a	Passed
LEGS	WRC-107/537	0.781	n/a	Passed
LEGS	WRC-107/537	0.502	n/a	Passed
N2-2"	WRC-107/537	0.692	n/a	Passed
N1-2"	WRC-297	0.775	Shell	Passed
N6-2"	WRC-297	0.775	Shell	Passed
N4-2"	WRC-107/537	0.692	n/a	Passed
N3-2"	WRC-107/537	0.692	n/a	Passed

Loads for Foundation/Support Design:

Total Wind Shear on top of all Legs	114. Kg
Total Earthquake Shear on top of all Legs	87. Kg
Total Wind Moment at top of all Legs	80. Kg-m.
Total Earthquake Moment at top of all Legs	97. Kg-m.
Wind Load Case Test Moment on Support	40. Kg-m.
Test Case Wind Shear on Support	57. Kg
Max. Wind Shear on one Leg (top & bottom)	45. Kg
Max. Earthq. Shear on one Leg (top & bottom)	35. Kg
Max. Wind Moment at base of one Leg	104. Kg-m.
Max. Earthquake Moment at base of one Leg	80. Kg-m.
Max. Vertical Load (Wt. + Wind) on one Leg	444. Kg
Max. Vertical Load (Wt. + Eq.) on one Leg	460. Kg

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	1619.9 kg.
Shop Test - Fabricated + water (full)	3451.7 kg.
Shipping - Fab. + rem. intls.+ shipping app.	1619.9 kg.
Erected - Fab. + rem. intls.+ insul. (etc)	1619.9 kg.
Empty - Fab. + intls. + details + wghts.	1619.9 kg.
Operating - Empty + operating liquid (no ca)	1622.3 kg.
Field Test - Empty weight + test medium (full)	3451.7 kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N2-2"	30.5	...	...	OK	5.02	OK	No Calc[*]
N1-2"	30.5	...	...	OK	5.02	OK	No Calc[*]
M1-6"	30.5	...	...	...		OK	Passed
N6-2"	30.5	...	...	OK	5.02	OK	No Calc[*]
K2-2"	30.5	...	...	OK	6.40	OK	No Calc[*]
K1A-3"	30.5	...	...	OK	7.33	OK	Passed
K1B-3"	30.5	...	...	OK	7.33	OK	Passed
N4-2"	30.5	...	...	OK	5.02	OK	No Calc[*]
N3-2"	30.5	...	...	OK	5.02	OK	No Calc[*]
N5-2"	30.5	...	...	OK	5.02	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 30.5 Nozzle : N5-2"
Minimum MAWP Shells/Flanges : 30.5 Element : Shell
Minimum MAPnc Shells/Flanges : 36.8 Element : Shell

Computed Vessel M.A.W.P. : 30.5 bars

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm.	Layout Angle deg	Dia. Limit mm.
10	N2-2"	0.000	0.000	96.468
20	N1-2"	1750.000	270.000	96.468
20	M1-6"	800.000	270.000	279.120
20	N6-2"	200.000	270.000	96.468
20	K2-2"	1800.000	90.000	113.200
20	K1A-3"	1600.000	90.000	158.800
20	K1B-3"	250.000	90.000	158.800
30	N4-2"	0.000	120.000	96.468
30	N3-2"	0.000	0.000	96.468
30	N5-2"	0.000	240.000	96.468

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N2-2"	2.000 in	160	WNF	2.375	8.738	...	...	216.50	600
N1-2"	2.000 in	160	WNF	2.375	8.738	...	...	216.91	600
N6-2"	2.000 in	160	WNF	2.375	8.738	...	...	216.91	600
K2-2"	2.000 in Actual	LWN	3.323	16.800	...	...	...	224.16	300
N4-2"	2.000 in	160	WNF	2.375	8.738	...	...	169.93	600
N3-2"	2.000 in	160	WNF	2.375	8.738	...	...	171.76	600
N5-2"	2.000 in	160	WNF	2.375	8.738	...	...	171.76	300
K1A-3"	3.000 in Actual	LWN	4.622	20.600	...	...	...	219.48	300
K1B-3"	3.000 in Actual	LWN	4.622	20.600	...	...	...	219.48	300
M1-6"	6.000 in	160	WNF	6.625	18.237	268.27	16.00	223.13	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N2-2"	SA-106 B	16.000	10.000	...	...	...
N1-2"	SA-106 B	16.000	10.000	...	...	...
N6-2"	SA-106 B	16.000	10.000	...	...	...
K2-2"	SA-105	16.000	10.000	...	...	...
N4-2"	SA-106 B	16.000	10.000	...	...	...
N3-2"	SA-106 B	16.000	10.000	...	...	...
N5-2"	SA-106 B	16.000	10.000	...	...	...
K1A-3"	SA-105	16.000	10.000	...	...	...
K1B-3"	SA-105	16.000	10.000	...	...	...
M1-6"	SA-106 B	16.000	10.000	12.000	16.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
N2-2"	...	0.0	200.00	0.00	bottom Dish
N1-2"	1750.000	270.0	200.00	0.00	Shell
N6-2"	200.000	270.0	200.00	0.00	Shell
K2-2"	1800.000	90.0	206.38	0.00	Shell
N4-2"	...	120.0	150.00	0.00	Top dish
N3-2"	...	0.0	150.00	0.00	Top dish
N5-2"	...	240.0	150.00	0.00	Top dish
K1A-3"	1600.000	90.0	200.02	0.00	Shell
K1B-3"	250.000	90.0	200.02	0.00	Shell
M1-6"	800.000	270.0	200.00	0.00	Shell

**Bill of Materials:**

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 20.0mm. THK X 1000.0mm. ID X 50.0mm.	SA-516 70
4	LEGS, SHAPE: ISA80X80X10 2300.0mm	...
4	BASEPLATES: 204.0mm X 204.0mm X 18.0mm	SA-516 70
4	BASEPLATE BOLTS: 24.0mm	SA-307 B
4	LEG RE-PADS: 150.000mm X 250.000mm X 16.000mm	SA-516 70
2	LIFTING LUGS:	SA-516 70
2	LIFTING LUG PADS: 380.0mm X 430.0mm X 16.0mm THK	
1	CLASS 300 GR 1.1, 6.0" BLIND FLANGE(S)	SA-105
1	CYLINDRICAL SEGMENT 16.0mm. THK X 3000.0mm. X 1900.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 16.0mm. THK X 141.6mm. X 1900.0mm.	SA-516 70
1	NAMEPLATE	...

**Minimum Design Metal Temperature Results Summary (°C):**

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
bottom Dish	[10]	B	-14	-27	-29	0.769	16.000	0.85	No
bottom Dish	[7]	B	-7	-29	-29	0.612	20.000	0.85	No
Shell	[8]	B	-14	-26	-29	0.782	16.000	0.85	No
Top dish	[10]	B	-14	-27	-29	0.769	16.000	0.85	No
Top dish	[7]	B	-7	-29	-29	0.612	20.000	0.85	No
N2-2"	[1]	B	-8	-104	-29	0.128	7.645	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N1-2"	[1]	B	-8	-104	-29	0.128	7.645	1.00	No
Nozzle Flg	[5]	A	-18	-104					
M1-6"	[1]	B	-14	-29	-29	0.780	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-40					
N6-2"	[1]	B	-8	-104	-29	0.128	7.645	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K2-2"	[1]	B	-14	-29	-29	0.780	16.000	1.00	No
Nozzle Flg	[5]	B	-18	-40					
K1A-3"	[1]	B	-14	-29	-29	0.780	16.000	1.00	No
Nozzle Flg	[5]	B	-18	-40					
K1B-3"	[1]	B	-14	-29	-29	0.780	16.000	1.00	No
Nozzle Flg	[5]	B	-18	-40					
N4-2"	[1]	B	-8	-104	-29	0.128	7.645	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N3-2"	[1]	B	-8	-104	-29	0.128	7.645	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N5-2"	[1]	B	-8	-104	-29	0.128	7.645	1.00	No
Nozzle Flg	[5]	A	-18	-40					
Nozl. Bolting[22]			-48			1.000	64.000	1.00	No

Warmest MDMT: -7 -26

Required Minimum Design Metal Temperature -12.0 °C

Warmest Computed Minimum Design Metal Temperature -29.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

MP Nitrogen Buffer Vessel(V8602)

PV Elite 28

Licensee: SL User



FileName : MP NITROGEN BUFFER VESSEL-V8602 7%

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MDMT Summary:

Step: 38 1:09pm Apr 20, 2026



HEXAGON

UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (bars):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

MP Nitrogen Buffer Vessel(V8602)

Design Internal Pressure (for Hydrotest)	30.5	bars
Design Internal Temperature	82.0	°C
Projection of Nozzle from Vessel Top	500	mm.
Projection of Nozzle from Vessel Bottom	500	mm.
Minimum Design Metal Temperature	-12.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-3	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE



Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	10000 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	2.000
Site Class	D
Component Elevation Ratio z/h	0.100
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

**Complete Listing of Vessel Elements and Details:**

Element#	1/3
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	bottom Dish
Distance "FROM" to "TO"	50 mm.
Inside Diameter	1000 mm.
Element Thickness	16 mm.
Internal Corrosion Allowance	1.6 mm.
Nominal Thickness	20 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	30.5 bars
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm.
Yield Stress, Operating	242.6 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	-250 mm.
Height/Length of Liquid	300 mm.
Liquid Density	0.12979E-05 kg./cm ³
Element From Node	10
Detail Type	Nozzle
Detail ID	N2-2"
Dist. from "FROM" Node / Offset dist	0 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	160
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element#	2/3
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	Shell
Distance "FROM" to "TO"	1900 mm.
Inside Diameter	1000 mm.
Element Thickness	16 mm.
Internal Corrosion Allowance	1.6 mm.
Nominal Thickness	16 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	30.5 bars
Design Temperature Internal Pressure	82 °C



Design External Pressure	0	bars
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	20	
Detail Type	Liquid	
Detail ID	Liquid: 20	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	1900	mm.
Liquid Density	0.12980E-05	kg./cm ³

Element From Node	20	
Detail Type	Nozzle	
Detail ID	N1-2"	
Dist. from "FROM" Node / Offset dist	1700	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	M1-6"	
Dist. from "FROM" Node / Offset dist	750	mm.
Nozzle Diameter	6	in.
Nozzle Schedule	160	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	N6-2"	
Dist. from "FROM" Node / Offset dist	150	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	K2-2"	
Dist. from "FROM" Node / Offset dist	1750	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	20	
Detail Type	Nozzle	



Detail ID	K1A-3"
Dist. from "FROM" Node / Offset dist	1550 mm.
Nozzle Diameter	3 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	20
Detail Type	Nozzle
Detail ID	K1B-3"
Dist. from "FROM" Node / Offset dist	200 mm.
Nozzle Diameter	3 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element From Node	20
Detail Type	Leg
Detail ID	LEGS
Dist. from "FROM" Node / Offset dist	250 mm.
Diameter at Leg Centerline	1120.3 mm.
Leg Orientation	3
Number of Legs	4
Section Identifier	ISA80X80X10
Length of Legs	2300 mm.

Element From Node	20
Detail Type	Weight
Detail ID	Clip weight
Dist. from "FROM" Node / Offset dist	0 mm.
Miscellaneous Weight	50 Kg
Offset from Element Centerline	0 mm.

Element#	3/3
Element From Node	30
Element To Node	40
Element Type	Elliptical
Description	Top dish
Distance "FROM" to "TO"	50 mm.
Inside Diameter	1000 mm.
Element Thickness	16 mm.
Internal Corrosion Allowance	1.6 mm.
Nominal Thickness	20 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	30.5 bars
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 mm.



Height/Length of Liquid	300	mm.
Liquid Density	0.11981E-05	kg./cm ³

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N4-2"	
Dist. from "FROM" Node / Offset dist	100	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	120.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N3-2"	
Dist. from "FROM" Node / Offset dist	150	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N5-2"	
Dist. from "FROM" Node / Offset dist	150	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	300	
Layout Angle	240.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

**XY Coordinate Calculations:**

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
bottom Dish		...	50	...	50
Shell		...	1950	...	1900
Top dish		...	2000	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
bottom Dish		30.5	20	1.6	1000	117.3	1
Shell		30.5	16	1.6	1000	117.3	2
Top dish		30.5	20	1.6	1000	117.3	3

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
bottom Dish		30.5	33.7	37.4	16	14.6222	1
Shell		30.5	33.1	36.8	16	14.8503	2
Top dish		30.5	33.7	37.4	16	14.6221	3

Minimum 30.5 36.8

MAWP: 30.5 bars, limited by: DESIGN (user specified)

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: bottom Dish

Elliptical Head Design Information:					
Design Pressure	P	30.500 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1000.000 mm.	Actual Stress		124.8 N./mm ²
Spec. Min. Thk t _e , or s _a		16.0000 mm.	Spec. Nominal Thk.		20.000 mm.
Surface Area		13358.0 cm ²	SG of Contents		0.001
Element Volume		170200.1 cm ³	Weight of Contents		0.2 Kgf
Empty Weight		217.9 Kgf	Operating Weight		218.1 Kgf
			Comp. All. Ope. Stress		137.3 N./mm ²
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 12.500 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.250 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (30.5 \cdot 1003 \cdot 1) / (2 \cdot 138 \cdot 0.85 - 0.2 \cdot 30.5)$$

$$= 13.0222 + 1.6000 = 14.6222 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.000 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 14.4) / (1 \cdot 1003 + 0.2 \cdot 14.4)$$

$$= 33.7 - 0.00032 = 33.7 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 16) / (1 \cdot 1000 + 0.2 \cdot 16)$$

$$= 37.4 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (30.5 \cdot (1 \cdot 1003 + 0.2 \cdot 14.4)) / (2 \cdot 0.85 \cdot 14.4)$$

$$= 124.829 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (30.5 \cdot 502) / (138 \cdot 0.85 - 0.6 \cdot 30.5) + 1.6$$

$$= 14.850 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.000 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 18.4) / (502 + 0.6 \cdot 18.4)$$

$$= 42.1 - 0.00028 = 42.1 \text{ bars}$$

Factor K, corroded condition [Kcor]:

$$\begin{aligned} &= (2 + (\text{Inside Diameter}/(2 \cdot \text{Inside Head Depth}))^2)/6 \\ &= (2 + (1003/(2 \cdot 252))^2)/6 \\ &= 0.995767 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 8.333 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 16$, $t_r = 13$, $c = 1.6$ mm., $E^* = 0.85$
Thickness Ratio = $t_r \cdot E^*/(t_g - c) = 0.77$, Temp. Reduction = 13 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-27 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 20$, $t_r = 13.3$, $c = 1.6$ mm., $E^* = 0.85$
Thickness Ratio = $t_r \cdot E^*/(t_g - c) = 0.61$, Temp. Reduction = 22 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-29 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	30.500 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	20.000 mm.	Basic MDMT	-7 °C
Stress/Thk. Ratio	0.612	Temperature Difference	22 °C
UCS-66(c) red.	No	Computed Min. Temp.	-29 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 12.500 mm.
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.250 mm.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Shell

Cylindrical Shell Design Information:					
Design Pressure	P	30.500 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		1900.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1000.000 mm.	Actual Stress		127.2 N./mm ²
Spec. Min. Thk t _e , or s _a		16.0000 mm.	Spec. Nominal Thk.		16.000 mm.
Surface Area		61600.3 cm ²	SG of Contents		0.001
Element Volume		1492523.8 cm ³	Weight of Contents		1.9 Kg
Empty Weight		752.1 Kg	Operating Weight		754.0 Kg
			Comp. All. Ope. Stress		137.3 N./mm ²
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		85 %	Long. Joint Eff.	E or v	85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (30.5 \cdot 502) / (138 \cdot 0.85 - 0.6 \cdot 30.5)$$

$$= 13.2503 + 1.6000 = 14.8503 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.000 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 14.4) / (502 + 0.6 \cdot 14.4)$$

$$= 33.1 - 0.00028 = 33.1 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 16) / (500 + 0.6 \cdot 16)$$

$$= 36.8 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (30.5 \cdot (502 + 0.6 \cdot 14.4)) / (0.85 \cdot 14.4)$$

$$= 127.152 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 1.575 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 16, t_r = 13.3, c = 1.6 mm., E* = 0.85

Thickness Ratio = t_r * E* / (t_g - c) = 0.78, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-26 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	30.500 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C

MP Nitrogen Buffer Vessel(V8602)

PV Elite 28 Licensee: SL User

FileName : MP NITROGEN BUFFER VESSEL-V8602 7%

Element: Shell



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HEXAGON

Governing Thk.	16.000 mm.	Basic MDMT	-14 °C
Stress/Thk. Ratio	0.782	Temperature Difference	12 °C
UCS-66(c) red.	No	Computed Min. Temp.	-26 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Top dish

Elliptical Head Design Information:					
Design Pressure	P	30.500 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.6000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1000.000 mm.	Actual Stress		124.8 N./mm ²
Spec. Min. Thk t _e , or s _a		16.0000 mm.	Spec. Nominal Thk.		20.000 mm.
Surface Area		13358.0 cm ²	SG of Contents		0.001
Element Volume		170200.1 cm ³	Weight of Contents		0.2 Kg _f
Empty Weight		217.9 Kg _f	Operating Weight		218.1 Kg _f
			Comp. All. Ope. Stress		137.3 N./mm ²
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 12.500 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.250 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (30.5 \cdot 1003 \cdot 1) / (2 \cdot 138 \cdot 0.85 - 0.2 \cdot 30.5)$$

$$= 13.0221 + 1.6000 = 14.6221 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.000 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 14.4) / (1 \cdot 1003 + 0.2 \cdot 14.4)$$

$$= 33.7 - 0.29406E-04 = 33.7 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 16) / (1 \cdot 1000 + 0.2 \cdot 16)$$

$$= 37.4 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (30.5 \cdot (1 \cdot 1003 + 0.2 \cdot 14.4)) / (2 \cdot 0.85 \cdot 14.4)$$

$$= 124.827 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (30.5 \cdot 502) / (138 \cdot 0.85 - 0.6 \cdot 30.5) + 1.6$$

$$= 14.850 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.000 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 18.4) / (502 + 0.6 \cdot 18.4)$$

$$= 42.1 - 0.29406E-04 = 42.1 \text{ bars}$$

Factor K, corroded condition [Kcor]:



$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (1003 / (2 \cdot 252))^2) / 6$$

$$= 0.995767$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 8.333 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 16$, $t_r = 13$, $c = 1.6$ mm., $E^* = 0.85$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.77$, Temp. Reduction = 13 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -27 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 20$, $t_r = 13.3$, $c = 1.6$ mm., $E^* = 0.85$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.61$, Temp. Reduction = 22 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -29 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	30.500 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	20.000 mm.	Basic MDMT	-7 °C
Stress/Thk. Ratio	0.612	Temperature Difference	22 °C
UCS-66(c) red.	No	Computed Min. Temp.	-29 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 12.500 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.250 mm.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	39.650 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	39.650 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	47.678 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	33.550 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	43.615 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	43.615 bars
App 27-4	= M.A.W.P.	30.500 bars

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
bottom Dish	176.2	234.0	0.753	47.78
Shell	179.0	234.0	0.765	47.78
Top dish	176.2	234.0	0.753	47.78

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N2-2"	Nozzle	118.00	118.00	1.000
N1-2"	Nozzle	118.00	118.00	1.000
M1-6"	Nozzle	118.00	118.00	1.000
M1-6"	Pad	138.00	138.00	1.000
N6-2"	Nozzle	118.00	118.00	1.000
K2-2"	Nozzle	138.00	138.00	1.000
K1A-3"	Nozzle	138.00	138.00	1.000
K1B-3"	Nozzle	138.00	138.00	1.000
N4-2"	Nozzle	118.00	118.00	1.000
N3-2"	Nozzle	118.00	118.00	1.000
N5-2"	Nozzle	118.00	118.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
bottom Dish	138.00	138.00	1.000
Shell	138.00	138.00	1.000
Top dish	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N2-2"	16.94	216.00	0.078
N1-2"	16.94	216.00	0.078
M1-6"	23.28	216.00	0.108
N6-2"	16.94	216.00	0.078
K2-2"	10.09	225.00	0.045
K1A-3"	11.70	225.00	0.052
K1B-3"	11.70	225.00	0.052
N4-2"	16.94	216.00	0.078
N3-2"	16.94	216.00	0.078
N5-2"	16.94	216.00	0.078

MP Nitrogen Buffer Vessel(V8602)

PV Elite 28 Licensee: SL User

FileName : MP NITROGEN BUFFER VESSEL-V8602 7%

Shop Pressure Test Results: Step: 7 1:09pm Apr 20,2026



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Elements Suitable for Test Pressure.

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**Hydrostatic Test Pressure Results:**

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	39.650 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	39.650 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	47.609 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	33.550 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	43.615 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	43.615 bars
App 27-4	= M.A.W.P.	30.500 bars

UG-99(b) Note 35, Test Pressure Calculation:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 30.5 • 1

= 39.650 bars

Vertical Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
bottom Dish	163.5	234.0	0.699	39.94
Shell	166.4	234.0	0.711	39.91
Top dish	162.6	234.0	0.695	39.73

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N2-2"	Nozzle	118.00	118.00	1.000
N1-2"	Nozzle	118.00	118.00	1.000
M1-6"	Nozzle	118.00	118.00	1.000
M1-6"	Pad	138.00	138.00	1.000
N6-2"	Nozzle	118.00	118.00	1.000
K2-2"	Nozzle	138.00	138.00	1.000
K1A-3"	Nozzle	138.00	138.00	1.000
K1B-3"	Nozzle	138.00	138.00	1.000
N4-2"	Nozzle	118.00	118.00	1.000
N3-2"	Nozzle	118.00	118.00	1.000
N5-2"	Nozzle	118.00	118.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
bottom Dish	138.00	138.00	1.000
Shell	138.00	138.00	1.000
Top dish	138.00	138.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N2-2"	14.16	216.00	0.066
N1-2"	14.15	216.00	0.066
M1-6"	19.45	216.00	0.090
N6-2"	14.15	216.00	0.066
K2-2"	8.43	225.00	0.037
K1A-3"	9.78	225.00	0.043

MP Nitrogen Buffer Vessel(V8602)

PV Elite 28 Licensee: SL User

FileName : MP NITROGEN BUFFER VESSEL-V8602 7%

Field Pressure Test Results: Step: 8 1:09pm Apr 20,2026



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K1B-3"	9.78	225.00	0.043
N4-2"	14.09	216.00	0.065
N3-2"	14.09	216.00	0.065
N5-2"	14.09	216.00	0.065

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	No Calc	1032	14.4	0.001938	103.041
20	30	2166.67	1032	14.4	0.0010214	85.9483
30	40	No Calc	1032	14.4	0.001938	103.041

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	16	No Calc	...	16
20	30	16	No Calc	...	16
30	40	16	No Calc	...	16

Minimum

16

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	2166.67	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm³	Corroded Metal Wgt. kg.	Corroded ID Volume cm³	Extra due Misc % kg.
10	20	217.855	170200	200.427	171713	15.2499
20	30	752.05	1492523	677.911	1502091	52.6435
30	40	217.855	170200	200.427	171713	15.2499
Total		1187	1832924.12	1078	1845516.75	83

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset, Dtl. Cent. mm.	Description
10	Liqd	0.22113	...	-125	...	Liquid: 10
10	Nozl	7.60088	...	-392	...	N2-2"
20	Liqd	1.93922	...	950	...	Liquid: 20
20	Nozl	7.60088	-600	1700	...	N1-2"
20	Nozl	64.771	-600	750	...	M1-6"
20	Nozl	7.60088	-600	150	...	N6-2"
20	Nozl	11.1574	603.187	1750	...	K2-2"
20	Nozl	19.9484	600.012	1550	...	K1A-3"
20	Nozl	19.9484	600.012	200	...	K1B-3"
20	Legs	139.934	...	-900	...	LEGS
20	Wght	50	...	...	...	Clip weight
30	Liqd	0.20412	...	175	...	Liquid: 30
30	Nozl	7.07638	86.6025	361.955	50	N4-2"
30	Nozl	7.07638	...	355.491	-150	N3-2"
30	Nozl	6.28336	-129.904	355.491	75	N5-2"

Total Weight of Each Detail Type:

Liquid	2.4
Nozzles	159.1
Legs	139.9
Weights	50.0
Sum of the Detail Weights	351.4 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	1270.9	1270.9	1270.9	1270.9	1270.9	1270.9
Nozzles	159.1	159.1	159.1	159.1	159.1	159.1
Legs	139.9	139.9	139.9	139.9	139.9	139.9
Wld Weights	50.0	50.0	50.0	50.0	50.0	50.0
Ope. Liquid	...	...	...	...	...	2.4
Test Liquid	...	1831.8	...	...	...	...
Totals	1619.9	3451.7	1619.9	1619.9	1619.9	1622.3

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	1619.9 kg.
Shop Test Wt.	- Fabricated weight + water (full)	3451.7 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	1619.9 kg.



Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	1619.9 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	1619.9 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	1622.3 kg.
Field Test Wt.	- Empty weight + test medium (full)	3451.7 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	509.9 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	949.5 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	553.4 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	13358	12410.6
20	30	61600.3	59690.3
30	40	13358	12410.6

Total	88316.4	84511.6 cm ²
Total	95.1	91.0 ft ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	240.706	240.927	410.802	...	240.706
20	Legs	129.7	129.955	325.965	17.3153	129.7
Legs	30	856.021	857.705	2151.37	...	856.021
30	40	253.541	253.745	423.637	0.2034	253.541

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
10	20	...	...	...
20	Legs	-240.706	-240.927	-410.802
Legs	30	1109.56	1111.45	2575.01
30	40	253.541	253.745	423.637

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
10	20	...	...	...
20	Legs	17.3153	17.3153	17.3153
Legs	30	0.2034	0.2034	0.2034
30	40	0.2034	0.2034	0.2034



Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Rating Ope.	Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
N2-2"	95.63	102.00	82	600	GR 1.1	53.407	90.043	95.633	90.043
N1-2"	95.63	102.00	82	600	GR 1.1	53.407	90.043	95.633	90.043
M1-6"	47.82	51.00	82	300	GR 1.1	...	...	...	...
N6-2"	95.63	102.00	82	600	GR 1.1	53.407	90.043	95.633	90.043
K2-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
K1A-3"	47.82	51.00	82	300	GR 1.1	...	...	...	...
K1B-3"	47.82	51.00	82	300	GR 1.1	...	...	...	...
N4-2"	95.63	102.00	82	600	GR 1.1	53.407	90.043	95.633	90.043
N3-2"	95.63	102.00	82	600	GR 1.1	53.407	90.043	95.633	90.043
N5-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
Min Rating	47.817	51.000, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm.	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kg	MAWP PD bars	Maximum Pressure Pmax bars	Pass Fail
N2-2"	77.851	0.50	50	0	30.500	90.043	Pass
N1-2"	77.851	0.50	50	0	30.500	90.043	Pass
N6-2"	77.851	0.50	50	0	30.500	90.043	Pass
N4-2"	77.851	0.50	50	0	30.500	90.043	Pass
N3-2"	77.851	0.50	50	0	30.500	90.043	Pass

Evaluating Flange on Nozzle [N2-2"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(50) + 4 \cdot \text{abs}(0) \cdot 77.851 \leq \pi \cdot 77.851^3 \cdot ((95.633 - 30.500) + .500 \cdot 95.633)$$

$$= 0.80732E+03 \leq 0.17074E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N2-2" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(95.633, 0.500 \cdot 95.633 + 95.633 - 807.3 / (\pi \cdot 77.851^3))$$

$$= \min(95.633, 90.043)$$

$$= 90.043 \text{ bars}$$

Evaluating Flange on Nozzle [N1-2"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(50) + 4 \cdot \text{abs}(0) \cdot 77.851 \leq \pi \cdot 77.851^3 \cdot ((95.633 - 30.500) + .500 \cdot 95.633)$$

$$= 0.80732E+03 \leq 0.17074E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1-2" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(95.633, 0.500 \cdot 95.633 + 95.633 - 807.3 / (\pi \cdot 77.851^3))$$

$$= \min(95.633, 90.043)$$

$$= 90.043 \text{ bars}$$

Evaluating Flange on Nozzle [N6-2"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(50) + 4 \cdot \text{abs}(0) \cdot 77.851 \leq \pi \cdot 77.851^3 \cdot ((95.633 - 30.500) + .500 \cdot 95.633)$$

$$= 0.80732E+03 \leq 0.17074E+04 \text{ Check Passes}$$



Maximum Pressure for Flange on Nozzle N6-2" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(95.633, 0.500 \cdot 95.633 + 95.633 - 807.3 / (\pi \cdot 77.851^3))$$

$$= \min(95.633, 90.043)$$

$$= 90.043 \text{ bars}$$

Evaluating Flange on Nozzle [N4-2"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(50) + 4 \cdot \text{abs}(0) 77.851 \leq \pi \cdot 77.851^3 [(95.633 - 30.500) + .500 \cdot 95.633]$$

$$= 0.80732E+03 \leq 0.17074E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N4-2" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(95.633, 0.500 \cdot 95.633 + 95.633 - 807.3 / (\pi \cdot 77.851^3))$$

$$= \min(95.633, 90.043)$$

$$= 90.043 \text{ bars}$$

Evaluating Flange on Nozzle [N3-2"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(50) + 4 \cdot \text{abs}(0) 77.851 \leq \pi \cdot 77.851^3 [(95.633 - 30.500) + .500 \cdot 95.633]$$

$$= 0.80732E+03 \leq 0.17074E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N3-2" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(95.633, 0.500 \cdot 95.633 + 95.633 - 807.3 / (\pi \cdot 77.851^3))$$

$$= \min(95.633, 90.043)$$

$$= 90.043 \text{ bars}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 16.478 Hz.

The Natural Frequency for the Vessel (Ope...) is 16.4648 Hz.

The Natural Frequency for the Vessel (Filled) is 11.0044 Hz.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	[V] 97.2	Km/hr
Surface Roughness Category	C: Open Terrain	
Type of Surface	Moderately Smooth	
Base Elevation	10000	mm.
Percent Wind for Hydrotest	50.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment H or Hh	0	mm.
Distance Upwind of Crest Lh	0	mm.
Distance from Crest to the Vessel x	0	mm.
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
= $\min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q)/(1 + 1.7 \cdot gV \cdot I_{zbar})))$
= $\min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.23 \cdot 0.97)/(1 + 1.7 \cdot 3.4 \cdot 0.23)))$
= $\min(0.85, 0.91)$
= 0.850

Natural Frequency of Vessel (Operating)	16.465 Hz
Natural Frequency of Vessel (Empty)	16.478 Hz
Natural Frequency of Vessel (Test)	11.004 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.521
Structure Height to Diameter ratio	2.252
Height to top of Structure	2266.000 mm.

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:
Exposure Category: C from Table 26.11-1
α = 9.5 : Zg = 274320 mm.

Effective Height [z]:
= Centroid Height + Vessel Base Elevation
= 180 + 10000 = 10180 mm.
= 33.4 ft. Imperial Units

Velocity Pressure coefficient evaluated at height z [Kz]:
Because z (33.4 ft.) > 15 ft.
= $2.01(z/Zg)^{2/\alpha}$
= $2.01(33.4/900)^{2/9.5}$
= 1

Type of Hill: No Hill

Wind Directionality Factor [Kd]:
= 1, per Table 26.6-1



As there is No Hill Present: [Kzt]:

$K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:

$= (1 + K1 \cdot K2 \cdot K3)^2$

$= (1 + 0 \cdot 0 \cdot 0)^2$

$= 1$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$

$= \max(16, 0.00256 \cdot 1 \cdot 1 \cdot 1 \cdot 1 \cdot 60.4^2)$

$= 16 \text{ psf } [78.1] \text{ Kgs/m}^2$

Force on the first element [F]:

$= qz \cdot G \cdot Cf \cdot \text{WindArea}$

$= 16 \cdot 0.85 \cdot 0.52 \cdot 3.45$

$= 24.4 \text{ lbs. } [11.1] \text{ Kgf}$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
bottom Dish	10179.7	0.000	0.000	0.000	1.005	1.000	78.120
Shell	11250.0	0.000	0.000	0.000	1.026	1.000	78.120
Top dish	12336.3	0.000	0.000	0.000	1.046	1.000	78.120

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

- Cf - Correction factor for natural frequency
- D - Average outer insulated diameter of vessel, mm.
- Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
- Dr - Average outer insulated of top half of vessel, mm.
- f - Natural frequency of vibration (Hertz)
- f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(104)$
- L - Total height of structure mm.
- Lc - Total length of conical section(s) of vessel, mm.
- tb - Uncorroded plate thickness at bottom of vessel, mm.
- V30 - Design Wind Speed provided by user, Km/hr
- Vc - Critical wind velocity, Km/hr
- Vw - Maximum wind speed at top of structure, Km/hr
- W - Total corroded weight of structure, Kgf
- Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. Kgf
- Z - Maximum amplitude of vibration at top of vessel, mm.
- Dl - Logarithmic decrement (taken as 0.03 for Welded Structures)
- Vp - Vib. Chance, $\leq 0.32037\text{E-}06$ (High); $0.32037\text{E-}06 < 0.40047\text{E-}06$ (Probable)
- P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5

:: $0/2000 = 0$

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)

:: $[3.39/6.56^2] \cdot 10^4 = 786$ [Geometry Violation]

Compute the vibration possibility. If $Vp > 0.40047\text{E-}06$ no chance. [Vp]:

$= W / (L \cdot Dr^2)$

$= 1506 / (2000 \cdot 1032^2)$

$= 0.70686\text{E-}06$

Since Vp is $> 0.40047\text{E-}06$ no further vibration analysis is required !

Wind Loads on Masses/Equipment/Piping

ID	Wind Area	Elevation	Pressure	Force
----	-----------	-----------	----------	-------



	cm ²	mm.	Kgs/m ²	Kgf
Clip weight	0.00	10050.00	78.12	0.00

The Natural Frequency for the Vessel (Ope...) is 16.4648 Hz.

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	10179.7	1238.4	3206.42	78.12	12.1991
20	30	11250	1238.4	23529.6	78.12	89.5205
30	40	12336.3	1238.4	3206.42	78.12	12.1991

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	2.000
Site Class	D
Component Elevation Ratio z/h	0.100
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1\end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
= $C_t \cdot h_n^x$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
= $0.020(13.90.75)$
= 0.144 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
= (1/Natural Frequency) = (1/16.5)
= 0.061

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
= Minimum Value of ($1.7 \cdot 0.14$, 0.061) per 12.8.2
= 0.061

Loads on Building/Structure Supported Equipment:

Seismic Force per equation 13.3-1, [Fp]:
= $0.4 \cdot A_p \cdot Sds \cdot W(1 + 2 \cdot (z/h)) / (R / I_e)$
= $0.4 \cdot 1 \cdot 0.16 \cdot 1482 (1 + 2 \cdot 0.1) / (2/1.25)$
= 69.729 Kgf

Maximum value of Fp per equation 13.3-2:
= $1.6 \cdot Sds \cdot I \cdot W$
= $1.6 \cdot 0.16 \cdot 1.25 \cdot 1482 = 465$ Kgf

Minimum value of Fp per equation 13.3-3:
= $0.3 \cdot 0.16 \cdot 1.25 \cdot 1482$
= 87.161 Kgf

Total Base Shear $V = F_p$, [V]:
= 87.161 Kgf

Vertical load per 12.4-4, [Ev]:
= $0.2 \cdot Sds \cdot W$
= $0.2 \cdot 0.16 \cdot 1482$
= 46.5 Kgf



Distribute the Base shear force to each element according to the equations

$F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor

$C_{vx} = W_x * h_{xk} / (\text{Sum of } W_i * h_{ik})$ and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1

The Natural Frequency for the Vessel (Ope...) is 16.4648 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf	Element Emp Load Kgf	Element Vertical Load Kgf
10	20	25	240.927	0.34743	0.3471	-7.55548
20	Legs	300	129.955	2.24881	2.24435	-11.6309
Legs	30	1125	857.705	55.658	55.5476	34.8551
30	40	1975	253.745	28.907	28.8831	7.95746



The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kg-f	Earthquake Shear Kg-f	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	386.265	...	...	...	...
20	Legs	125	12.1991	0.34743	1.66234	0.047343
Legs	30	825	113.919	87.1612	79.7447	97.1399
30	40	1786.26	12.1991	28.907	1.66234	3.93908

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
Wind: 1.100; Earthquake: 1.000.

Note:

Review the Vessel Design Summary for the cumulative shear force and bending moment on the support.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear Kg	Centroid Deflection mm.	Elem. End Deflection mm.	Elem. Ang. Rotation
10	20	...	6.34995	6.34995	0.0041411
20	Legs	12.1991	6.34996	6.34997	0.0041413
Legs	30	113.919	6.35035	6.35092	0.0041418
30	40	12.1991	6.35094	6.35096	0.0041418

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed Km/hr	2nd Crit. Wind Speed Km/hr
10	20	366.032	2287.7
20	30	366.032	2287.7
30	40	366.032	2287.7

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 10.000 Actual deflection : 6.351 mm.

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**Longitudinal Stress Constants:**

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm. ³	Section Modulus Corroded mm. ³
10	20	510.697	460.352	12773759	11550243
20	30	510.697	460.352	12773759	11550243
30	40	510.697	460.352	12773759	11550243

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**Longitudinal Allowable Stresses:**

From	To	Tensile N./mm ²	Hydrotest Tensile N./mm ²	Compressive N./mm ²	Hydrotest Compressive N./mm ²
10	20	140.757	238.676	-137.327	-137.327
20	Legs	140.757	238.676	-137.327	-137.327
Legs	30	140.757	238.676	-137.327	-137.327
30	40	140.757	238.676	-137.327	-137.327

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure N./mm ²	Longitudinal Stress External Pressure N./mm ²	Longitudinal Stress Hydrotest Pressure N./mm ²
10	20	52.5139	...	68.2681
20	30	52.5139	...	68.2681
30	40	52.5139	...	68.2681

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty N./mm ²	Wght. Str. Operating N./mm ²	Wght. Str. Hydrotest N./mm ²	Wght. Str. Emp. Mom. N./mm ²	Wght. Str. Opr. Mom. N./mm ²
10	20	...	...	...	...	...
20	Legs	0.051277	0.051324	0.087513	0.014701	0.014701
Legs	30	-0.23637	-0.23631	-0.23637	0.0001727	0.0001727
30	40	-0.054012	-0.054012	-0.054012	0.0001727	0.0001727

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. N./mm ²	Bend. Str. Oper. Wind N./mm ²	Bend. Str. Oper. Equ. N./mm ²	Bend. Str. Hyd. Wind N./mm ²	Bend. Str. Hyd. Equ. N./mm ²
10	20	...	...	...	...	...
20	Legs	0.014701	0.0014114	0.40196E-04	0.0007057	...
Legs	30	0.0001727	0.067707	0.082476	0.033853	...
30	40	0.0001727	0.0014114	0.0033444	0.0007057	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case N./mm ²	Vortex Shedding Empty Case N./mm ²	Vortex Shedding Test Case N./mm ²	Earthquake Empty Case N./mm ²
10	20	...	...	...	...
20	Legs	...	...	...	0.40158E-04
Legs	30	...	...	...	0.082363
30	40	...	...	...	0.0033417

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind N./mm ²	Longitudinal Stress Y Forces Seismic N./mm ²
10	20	...	-0.0016095
20	Legs	...	-0.0024777
Legs	30	...	0.0074251
30	40	...	0.0016952

Longitudinal Stresses due to User Forces and Moments:



From	To	Wind For/Mom Corroded N./mm ²	Earthquake For/Mom Corroded N./mm ²	Wind For/Mom No Corrosion N./mm ²	Earthquake For/Mom No Corrosion N./mm ²
10	20	...	...	...	...
20	Legs	...	...	...	...
Legs	30	...	...	...	...
30	40	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress N./mm ²
10	20	...
20	Legs	...
Legs	30	...
30	40	...



Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	140.76	0.00	137.33	0.0000	-0.0000
20	0.07	140.76		137.33	0.0005	
20		140.76	-0.30	137.33		0.0022
30		140.76	-0.06	137.33		0.0004

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		140.76	-0.00	137.33		0.0000
20	0.06	140.76		137.33	0.0005	
20		140.76	-0.31	137.33		0.0023
30		140.76	-0.06	137.33		0.0004

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	140.76	0.00	137.33	0.0000	-0.0000
20	0.07	140.76		137.33	0.0005	
20		140.76	-0.30	137.33		0.0022
30		140.76	-0.06	137.33		0.0004



Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		140.76	-0.00	137.33		0.0000
20	0.06	140.76		137.33	0.0005	
20		140.76	-0.31	137.33		0.0023
30		140.76	-0.06	137.33		0.0004

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	238.68	0.00	137.33	0.0000	-0.0000
20	0.10	238.68		137.33	0.0004	
20		238.68	-0.27	137.33		0.0020
30		238.68	-0.05	137.33		0.0004

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	238.68	0.00	137.33	0.0000	-0.0000
20	0.10	238.68		137.33	0.0004	
20		238.68	-0.24	137.33		0.0017
30		238.68	-0.05	137.33		0.0004

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	52.51	140.76		137.33	0.3731	
20	52.58	140.76		137.33	0.3736	
20	52.35	140.76		137.33	0.3719	
30	52.46	140.76		137.33	0.3727	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	52.51	140.76		137.33	0.3731	
20	52.58	140.76		137.33	0.3735	
20	52.37	140.76		137.33	0.3720	
30	52.47	140.76		137.33	0.3727	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	140.76	0.00	137.33	0.0000	-0.0000
20	0.07	140.76		137.33	0.0005	
20		140.76	-0.30	137.33		0.0022
30		140.76	-0.06	137.33		0.0004

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		140.76	-0.00	137.33		0.0000
20	0.06	140.76		137.33	0.0005	
20		140.76	-0.31	137.33		0.0023
30		140.76	-0.06	137.33		0.0004

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	68.27	238.68		137.33	0.2860	
20	68.37	238.68		137.33	0.2865	



20	68.07	238.68	137.33	0.2852
30	68.21	238.68	137.33	0.2858

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	68.27	238.68		137.33	0.2860	
20	68.37	238.68		137.33	0.2865	
20	68.03	238.68		137.33	0.2850	
30	68.21	238.68		137.33	0.2858	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	52.51	140.76		137.33	0.3731	
20	52.58	140.76		137.33	0.3735	
20	52.28	140.76		137.33	0.3714	
30	52.46	140.76		137.33	0.3727	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	52.51	140.76		137.33	0.3731	
20	52.56	140.76		137.33	0.3734	
20	52.30	140.76		137.33	0.3716	
30	52.47	140.76		137.33	0.3727	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	52.51	140.76		137.33	0.3731	
20	52.58	140.76		137.33	0.3735	
20	52.28	140.76		137.33	0.3714	
30	52.46	140.76		137.33	0.3727	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	52.51	140.76		137.33	0.3731	
20	52.58	140.76		137.33	0.3735	
20	52.28	140.76		137.33	0.3714	
30	52.46	140.76		137.33	0.3727	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.00	140.76	0.00	137.33	0.0000	-0.0000
20	0.07	140.76		137.33	0.0005	
20		140.76	-0.24	137.33		0.0017
30		140.76	-0.05	137.33		0.0004

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	52.51	117.30		114.44	0.4477	
20	52.58	117.30		114.44	0.4482	
20	52.29	117.30		114.44	0.4457	
30	52.46	117.30		114.44	0.4473	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
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10		117.30	-0.00	114.44		0.0000
20	0.06	117.30		114.44	0.0005	
20		117.30	-0.23	114.44		0.0020
30		117.30	-0.05	114.44		0.0005

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	52.51	140.76		137.33	0.3731	
20	52.58	140.76		137.33	0.3735	
20	52.28	140.76		137.33	0.3714	
30	52.46	140.76		137.33	0.3727	

Absolute Maximum of the all of the Stress Ratio's 0.4482

Governing Element: Shell

Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
bottom Dish	52.5	...	...	...	52.5	+ Tens
bottom Dish	52.5	...	...	...	...	- Comp
Shell	52.5	...	...	...	52.6	+ Tens
Shell	52.5	...	...	...	...	- Comp
Shell	52.5	-0.2	...	...	52.3	+ Tens
Shell	52.5	-0.2	...	...	...	- Comp
Top dish	52.5	-0.1	...	...	52.5	+ Tens
Top dish	52.5	-0.1	...	...	...	- Comp



Shop/Field Installation Options :

Note : The CG is computed from the first Element From Node

Center of Gravity of the Liquid	991.907 mm.
Center of Gravity of the Nozzles	1054.971 mm.
Center of Gravity of the Legs	-850.000 mm.
Center of Gravity of the Added Weights (Operating)	50.000 mm.
Center of Gravity of the Added Weights (Empty)	50.000 mm.
Center of Gravity of Bare Shell New and Cold	1000.000 mm.
Center of Gravity of Bare Shell Corroded	1000.000 mm.
Vessel CG in the Operating Condition	802.308 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	816.264 mm.
Vessel CG in the Test Condition	913.772 mm.

Rigging Analysis Results:

- Lifting weight based on the user selected calculated erected weight

Total Effective Length of Vessel for this analysis	2000.00 mm.
Total vessel weight (No Liquid) Twt	1619.90 Kgf
Impact weight multiplication factor Imp	1.50
Design lifting weight, DWT = Imp * Twt	2429.85 Kgf
Elevation of the Tailing Lug (bottom)	150.00 mm.
Elevation of the Lifting Lug (top)	1690.00 mm.
Design Reaction force at the tailing lug	1378.60 Kgf
Design Reaction force at the lifting lug	1051.25 Kgf
CG distance during field lifting operation	816.26 mm.
CG distance from tailing lug	666.26 mm.
CG distance from the nearer lifting lug/trunnion	666.26 mm.
Compressive Allowable Increase Factor	1.00

Critical Values: (N./mm², mm.)

	Max Stress	Elevation	Allowables
Bending	0.34	1570.00	116.05 (UG-23)
Shear	0.34	10.00	96.60 (0.7*S)

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

Distance mm.	Bending Moment Kg-m.	Bending Stress N./mm ²	Shear Force Kgf	Shear Stress N./mm ²
0.00	0.0	0.0	1701.7	0.3
430.00	203.8	0.2	883.0	0.2
1190.00	324.6	0.2	1069.5	0.2
1950.00	207.4	0.2	204.3	0.0

Unity Check (Actual Stress / Allowable Stress):

Maximum Unity Check is 0.0029 at elevation 1570 mm. - Must be <=1

Note: The rigging analysis is performed using a uniformly distributed load.

--- Plot data successfully generated ---

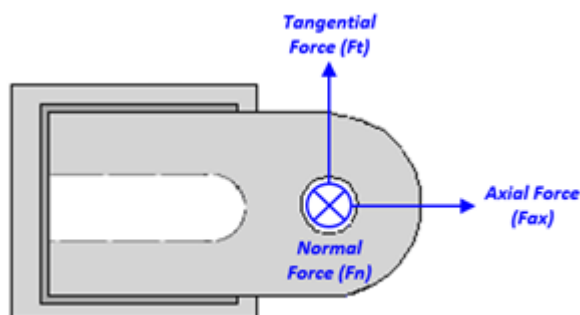


Lifting Lug Calculations:

Input Values:

Lifting Lug Material		SA-516 70
Lifting Lug Yield Stress	Yield	262.00 N./mm ²
Width of Lifting Lug	w	330.0000 mm.
Thickness of Lifting Lug	t	20.0000 mm.
Diameter of Hole in Lifting Lug	dh	50.0000 mm.
Radius of Semi-Circular Arc of Lifting Lug	r	100.0000 mm.
Height of Lug from Bottom to Center of Hole	h	900.0000 mm.
Offset from Vessel OD to Center of Hole	off	15.0000 mm.
Lug Fillet Weld Size	tw	16.0000 mm.
Length of weld along side of Lifting Lug	wl	380.0000 mm.
Length of Weld along Bottom of Lifting Lug	wb	330.0000 mm.
Thickness of Collar (if any)	tc	0.0000 mm.
Diameter of Collar (if any)	dc	0.0000 mm.
Impact Factor	Impfac	1.50
Number of Lugs in Group		2
Reinforcing Pad Circumferential Width	wpb	380.00 mm.
Reinforcing Pad Longitudinal Length	wpl	430.00 mm.
Reinforcing Pad Thickness	thkp	16.00 mm.
Reinforcing Pad Fillet Weld Size	tfp	10.00 mm.

Lifting Lug Orientation to Vessel: Flat
Lift Orientation : From Horizontal to Vertical



PV Elite computes the forces and stresses from 0 to 90 degrees at 1 degree increments. The table below provides the results in 5 degree increments.

The 0 degree angle position is in the horizontal direction and the 90 degree angle position is in the vertical direction.

PV Elite does not compute weak axis bending forces on the lugs. It is assumed that a spreader bar is used.

Lifting Lug and Weld Stresses at various Angles: N./mm²

Angle	Weld Stresses			Lug Stresses		Maximum Ratio
	Primary	Secondary	Allowable	Combined	Allowable	
0	0.5	2.9	104.8	7.4	172.9	0.043
5	0.5	3.0	104.8	7.7	172.9	0.045
10	0.5	3.1	104.8	8.0	172.9	0.046
15	0.6	3.1	104.8	8.2	172.9	0.048
20	0.6	3.2	104.8	8.4	172.9	0.048
25	0.6	3.1	104.8	8.4	172.9	0.049
30	0.6	3.1	104.8	8.4	172.9	0.049
35	0.7	3.0	104.8	8.3	172.9	0.048
40	0.7	2.9	104.8	8.2	172.9	0.047



45	0.7	2.8	104.8	7.9	172.9	0.046
50	0.7	2.6	104.8	7.6	172.9	0.044
55	0.8	2.4	104.8	7.2	172.9	0.041
60	0.8	2.2	104.8	6.7	172.9	0.039
65	0.8	1.9	104.8	6.1	172.9	0.035
70	0.9	1.7	104.8	5.5	172.9	0.032
75	0.9	1.3	104.8	4.7	172.9	0.027
80	1.0	1.0	104.8	3.9	172.9	0.023
85	1.1	0.5	104.8	2.9	172.9	0.017
90	1.2	0.1	104.8	1.8	172.9	0.011

Computed Results:

Total vessel weight (no liquid) 1619.90 Kgf
 Design reaction force at the tailing lug 1378.60 Kgf
 Design reaction force at the lifting lug 1051.25 Kgf

Worst case lift angle for primary weld stresses 90 deg
 Tangential force at this angle 0.0 Kgf
 Axial force at this angle 1214.9 Kgf

Worst case lift angle for secondary weld stresses 21 deg
 Tangential force at this angle 574.6 Kgf
 Axial force at this angle 220.6 Kgf

Worst case lift angle for lug stresses 26 deg
 Tangential force at this angle 571.6 Kgf
 Axial force at this angle 278.8 Kgf

Converting the weld leg dimension (tw) to the weld throat dimension.

Weld Group Inertia Calculations:

Weld group inertia about the longitudinal axis I_{ll} 20313.006 cm⁴
 Weld group centroid distance in the long. direction Y_{ll} 249.235 mm.
 Dist. of weld group centroid from lug bottom Y_{llb} 130.765 mm.
 Weld group inertia about the circumferential axis I_{lc} 25225.332 cm⁴
 Weld group centroid distance in the circ. direction Y_{lc} 165.000 mm.

Note: The Impact Factor is applied to the Forces acting on the Lug.

Primary Shear Stress in the Welds due to Shear Loads [S_{sl}]:

$$= \sqrt{(F_{ax}^2 + F_t^2 + F_n^2)} / ((2 \cdot w_l + w_b) \cdot t_w)$$

$$= \sqrt{(1215^2 + 0^2 + 0^2)} / ((2 \cdot 380 + 330) \cdot 11.3)$$

$$= 0.97 \text{ N./mm}^2$$

Shear Stress in the Welds due to Bending Loads [S_{blf}]:

$$= (F_n(h - Y_{llb}))Y_{ll}/I_{ll} + (F_{ax} \cdot \text{off} \cdot Y_{ll}/I_{ll}) + (F_t \cdot \text{off} \cdot Y_{lc}/I_{lc})$$

$$= (0(900 - 131))249/20313 +$$

$$(1215 \cdot 15 \cdot 249/20313) +$$

$$(0 \cdot 15 \cdot 165/25225)$$

$$= 0.22 \text{ N./mm}^2$$

Total Shear Stress for Combined Loads [S_t]:

$$= S_{sl} + S_{blf}$$

$$= 0.97 + 0.22$$

$$= 1.19 \text{ N./mm}^2$$

Allowable Shear Stress for Combined Loads [S_{ta}]:

$$= 0.4 \cdot \text{Yield} \cdot \text{Occfac (AISC Shear Allowable)}$$

$$= 0.4 \cdot 262 \cdot 1$$

$$= 104.80 \text{ N./mm}^2$$

Secondary Shear Stress in the Welds due to Shear Loads:

Unit Weld Polar Section Modulus [U_{wsm}]:

$$= (2 \cdot w_l + w)^3 / 12 - w_l^2(w_l + w)^2 / (2 \cdot w_l + w)$$

$$= 41137404.00 \text{ mm}^3$$

**Loads on Welds due to Torsional Moment:**

Shear Stress in Normal Direction [Fth]:

$$= F_t \cdot (h - (w_l - \text{Cent})) \cdot (w_b / 2) / U_{wsm}$$

$$= 1.77 \text{ Kg/mm.}$$

Shear Stress in Circumferential Direction [Ftv]:

$$= F_t \cdot (h - (w_l - \text{Cent})) \cdot \text{Cent} / U_{wsm}$$

$$= 2.65 \text{ Kg/mm.}$$

Primary Shear Stress [Fsv]:

$$= F_t / (2 \cdot w_l + w)$$

$$= 0.53 \text{ Kg/mm.}$$

Resultant Load on Weld Group [Fr]:

$$= \sqrt{(F_{th}^2 + (F_{tv} + F_{sv})^2)}$$

$$= \sqrt{(1.77^2 + (2.65 + 0.53)^2)}$$

$$= 3.64 \text{ Kg/mm.}$$

Resultant Secondary Weld Stress [Fws]:

$$= F_r / T_{weld}$$

$$= 3.64 / 11.3$$

$$= 3.16 \text{ N./mm}^2$$

Allowable Resultant Secondary Weld Stress [Psa]:

$$= 0.4 \cdot \text{Yield} \cdot \text{Occfac}$$

$$= 0.4 \cdot 262 \cdot 1$$

$$= 104.80 \text{ N./mm}^2$$

Shear Stress in Lug above Hole [Shs]:

$$= \sqrt{(P_l^2 + F_n^2)} / S_{ha}$$

$$= \sqrt{(1215^2 + 0^2)} / 30$$

$$= 3.97 \text{ N./mm}^2$$

Allowable Shear Stress in Lug above Hole [Sas]:

$$= 0.4 \cdot \text{Yield} \cdot \text{Occfac}$$

$$= 0.4 \cdot 262 \cdot 1$$

$$= 104.80 \text{ N./mm}^2$$

Pin Hole Bearing Width [Pbw]:

$$= \text{Lug Thickness} + 2 \cdot \text{Collar Thickness}$$

$$= 20 + 2 \cdot 0$$

$$= 20.000 \text{ mm.}$$

Pin Hole Bearing Stress [Pbs]:

$$= \sqrt{(P_l^2 + F_n^2)} / (P_{bw} \cdot d_h)$$

$$= \sqrt{(1215^2 + 0^2)} / (20 \cdot 50)$$

$$= 11.91 \text{ N./mm}^2$$

Allowable Bearing Stress [Pba]:

$$= \min(0.75 \cdot \text{Yield} \cdot \text{Occfac}, 0.9 \cdot \text{Yield}) \text{ AISC Bearing All.}$$

$$= \min(0.75 \cdot 262 \cdot 1, 236)$$

$$= 196.50 \text{ N./mm}^2$$

Bending Stress at the Base of the Lug [Fbs]:

$$= F_n \cdot (h - w_l) / (w \cdot t^2 / 6) + F_t \cdot (h - w_l) / (w^2 \cdot t / 6)$$

$$= 0 \cdot (900 - 380) / (330 \cdot 20^2 / 6) +$$

$$572 \cdot (900 - 380) / (330^2 \cdot 20 / 6)$$

$$= 8.03 \text{ N./mm}^2$$

Tensile Stress at the Base of the Lug [Fa]:

$$= F_{ax} / (w \cdot t)$$

$$= 279 / (330 \cdot 20)$$

$$= 0.41 \text{ N./mm}^2$$

Total Combined Stress at the Base of the Lug:

$$= F_{bs} + F_a$$

$$= 8.03 + 0.41$$



= 8.44 N./mm²

Lug Allowable Stress for Bending and Tension:

= min(0.66 • Yield • Occfac, 0.75•Yield)
 = min(0.66 • 262 • 1, 196)
 = 172.92 N./mm²

Required Shackles Pin Diameter [Spd]:

= $\sqrt{((2(Pl^2 + Fn^2)^{1/2}/(\pi \cdot Sta)))}$
 = $\sqrt{(2(1215^2 + 0^2)^{1/2}/(\pi \cdot 105))}$
 = 8.5075 mm.

WRC 107/537 Stress Analysis for the Lifting Lug to Shell Junction in the new and Cold Condition (no corrosion applied).

The full lug load is applied in both the circ. and long. directions simultaneously.

Attachment dimensions (C11P, C22P, etc.) may be modified accounting for attachment welds where applicable.

Input Echo, WRC107/537 Item 1, Description: Lift Lug

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Cylindrical
Internal Corrosion Allowance	Cas	0.0000 mm.
Vessel Diameter	Dv	1000.000 mm.
Vessel Thickness	Tv	16.000 mm.

Note: Lifting lug calculations are performed at ambient temperature conditions.

Attachment Type	Type	Rectangular
Parameter C11	C11	362.00 mm.
Parameter C22	C22	396.00 mm.
Thickness of Reinforcing Pad	Tpad	16.000 mm.
Pad Parameter C11P	C11p	400.000 mm.
Pad Parameter C22P	C22p	440.000 mm.
Design Internal Pressure	Dp	0.000 bars
Include Pressure Thrust		No

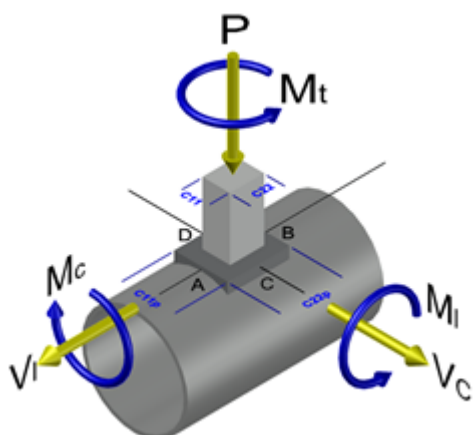
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	0.0 Kg
Longitudinal Shear (SUS)	Vl	1214.9 Kg
Circumferential Shear (SUS)	Vc	1214.9 Kg
Circumferential Moment (SUS)	Mc	0.0 Kg-m.
Longitudinal Moment (SUS)	ML	0.0 Kg-m.
Torsional Moment (SUS)	Mt	862.6 Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

WRC 107 Stress Calculation for SUSained loads:

Radial Load	P	0.0	Kgf
Circumferential Shear	VC	1214.9	Kgf
Longitudinal Shear	VL	1214.9	Kgf
Circumferential Moment	MC	0.0	Kg-m.
Longitudinal Moment	ML	0.0	Kg-m.
Torsional Moment	MT	862.6	Kg-m.

Dimensionless Parameters used : Gamma = 16.12

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.387	4C	2.024	(A,B)
N(PHI) / (P/Rm)	0.387	3C	1.241	(C,D)
M(PHI) / (P)	0.362	2C1	0.026	(A,B)
M(PHI) / (P)	0.362	1C !	0.056	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.361	3A	0.685	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.405	1A	0.072	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.372	3B	1.516	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.381	1B	0.019	(A,B,C,D)
N(x) / (P/Rm)	0.375	3C	1.284	(A,B)
N(x) / (P/Rm)	0.375	4C	2.067	(C,D)
M(x) / (P)	0.377	1C1	0.046	(A,B)
M(x) / (P)	0.377	2C !	0.030	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.361	4A	1.524	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.452	2A	0.032	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.372	4B	0.666	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.399	2B	0.031	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Circ. Bend. MC		0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0

Circ. Memb. ML	-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Bend. ML	-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Circ. Str.	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Memb. P	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P	-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Long. Bend. MC	0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Long. Memb. ML	-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Bend. ML	-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Long. Str.	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear VC	0.5	0.5	-0.5	-0.5	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Shear MT	7.4	7.4	7.4	7.4	7.4	7.4	7.4	7.4
Tot. Shear	7.9	7.9	6.9	6.9	6.9	6.9	7.8	7.8
Str. Int.	15.8	15.8	13.7	13.7	13.8	13.8	15.7	15.7

Dimensionless Parameters used : Gamma = 31.75

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.437	4C	2.801	(A,B)
N(PHI) / (P/Rm)	0.437	3C	1.178	(C,D)
M(PHI) / (P)	0.407	2C1	0.011	(A,B)
M(PHI) / (P)	0.407	1C !	0.061	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.406	3A	0.996	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.452	1A	0.064	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.420	3B	1.705	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.428	1B	0.009	(A,B,C,D)
N(x) / (P/Rm)	0.423	3C	1.236	(A,B)
N(x) / (P/Rm)	0.423	4C	2.888	(C,D)
M(x) / (P)	0.425	1C1	0.024	(A,B)
M(x) / (P)	0.425	2C !	0.031	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.406	4A	3.105	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.502	2A !	0.026	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.420	4B	0.974	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.447	2B	0.014	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Circ. Bend. MC		0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Circ. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Circ. Str.		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0



Circ. Q (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Q (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)	0.9	0.9	-0.9	-0.9	-0.8	-0.8	0.8	0.8
Shear Q (SUS)	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0
Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)	1.9	1.9	1.9	1.9	1.7	1.7	1.7	1.7
Pm+Pl+Q (Total)	25.9	25.9	22.2	22.2	22.3	22.3	25.7	25.7

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	1.86	207.00	Passed
Pm+Pl+Q (TOTAL)	25.89	413.99	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

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RESULTS FOR LEGS : Operating Case Description: LEGS

Legs attached to: Shell

Section Properties : Single Angle ISA80X80X10

India ISI Structural Steel Data

Overall Leg Length		2300.000	mm.
Effective Leg Length	Leglen	2000.000	mm.
Distance Leg Up Side of Vessel		250.000	mm.
Number of Legs	Nleg	4	
Cross Sectional Area for ISA80X80X10	Aleg	15.050	cm ²
Section Inertia (strong axis)		87.700	cm ⁴
Section Inertia (weak axis)		87.700	cm ⁴
Section Modulus (strong axis)		15500.063	mm. ³
Section Modulus (weak axis)		15500.063	mm. ³
Radius of Gyration (strong axis)		24.100	mm.
Radius of Gyration (weak axis)		24.100	mm.

Leg Orientation - Diagonal

Overturning Moment at top of Legs		97.1	Kg-m.
Total Weight Load at top of Legs	W	1491.5	Kgf
Total Shear force at top of Legs		113.9	Kgf
Additional force in Leg due to Bracing	Fadd	0.0	Kgf
Occasional Load Factor	Occfac	1.333	
Effective Leg End Condition Factor	k	1.000	

Note: The Legs are Not Cross Braced

The Leg Shear Force includes Wind, Seismic, and Applied Load Effects.

Pad Width along Circumference	C11P	150.000	mm.
Pad Length along Vessel Axis	C22P	250.000	mm.
Pad Thickness	Tpad	16.000	mm.

Maximum Shear at top of one Leg [Vleg]:
 $= (\max(\text{Wind, Seismic, Blast}) + \text{applied forces}) (I_{\max} / I_{\text{tot}})$
 $= (114) (141/356)$
 $= 45.20 \text{ Kgf}$

Axial Compression, Leg furthest from the Neutral Axis [Sma]:
 $= (W/Nleg + (Mleg / (Nlegm \cdot Rn))) / Aleg$
 $= (14626/4 + (952542 / (2 \cdot 560))) / 1505$
 $= 2.99 \text{ N./mm}^2$

Axial Compression, Leg closest to the Neutral Axis [Sva]:
 $= (W / Nleg) / Aleg$
 $= (1491/4) / 15$
 $= 2.43 \text{ N./mm}^2$

Computing Principal Axis and Inertias for Angle:

Leg lengths and thickness:	80.000	80.000	10.000
Distance to geometric centroid:	23.400	23.400	
Arm about YY:	18.400	21.600	
Arm about ZZ:	16.600	18.400	
Leg areas:	8.0000	7.0000	
Geometric inertia components YY:	27.751	61.242	
Geometric inertia components ZZ:	64.711	24.282	
Geometric inertias Iy & Iz:	88.993	88.993	
Product of inertia:	52.256		
Mohrs Radius:	52.256		
Average Inertia:	88.993		

QFACT = 1.0000 FBZ = 0.16383
 Principal Axis Inertias (Z&W) = 36.738 141.25



Angle to Principal Axis = 45.000
 Distances to extreme fibers CW & CZ = 56.568 23.476
 FOB from Eq 5-5 = 0.97392
 Bending allowables Fby & Fbz = 0.16383 0.16383

Shear Center Coordinates Wo & Zo: 25.095 0.0000

Values for Elastic Flexural-Torsional Buckling Stress:

E, G, J, R0²: 29500. 11346. 0.12053 2.8092
 AREA, LENGTH, Kw, Kz: 2.3328 78.740 1.0000 1.0000
 H, Few, Fez, Fej: 0.65252 68.314 17.768 208.68
 Fe computed from C4-1: 59.926

Initial (Kl/r)_{max}, & (Kl/r)_{equiv} = 128.01 69.704
 Final (Kl/r)_{max}, & Cc = 128.01 127.18
 Fa based on Eq 4-2 = 0.63918E-01

Leg Bending and Axial Stresses vs. Allowables: (N./mm²)

	Actual	Allowable
Weak Axis Bending :	40.06	218.38
Strong Axis Bending :	25.11	218.38
Axial Compression :	2.99	85.20

UNITY CHECKS ARE: H1-1 0.000
 H1-2 0.000
 H1-3 0.334

AISC Unity Check : 0.334, should be <= to 1 for Code compliance, [Passed]

WRC 107 Stress Analysis for Leg to Shell Junction, Ope Condition

Rectangular Attachment Parameter C11 113.137 mm.
 Rectangular Attachment Parameter C22 230.950 mm.

Input Echo, WRC107/537 Item 1, Description: LEGS

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Cylindrical	
Internal Corrosion Allowance	Cas	1.6000	mm.
Vessel Diameter	Dv	1000.000	mm.
Vessel Thickness	Tv	16.000	mm.
Design Temperature	T1	82.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Rectangular	
Parameter C11	C11	113.14	mm.
Parameter C22	C22	230.95	mm.
Thickness of Reinforcing Pad	Tpad	16.000	mm.
Pad Parameter C11P	C11p	150.000	mm.
Pad Parameter C22P	C22p	250.000	mm.
Design Internal Pressure	Dp	30.500	bars
Include Pressure Thrust		No	
Vessel Centerline Direction Cosine	Vx	0.000	
Vessel Centerline Direction Cosine	Vy	1.000	
Vessel Centerline Direction Cosine	Vz	0.000	
Nozzle Centerline Direction Cosine	Nx	1.000	



Nozzle Centerline Direction Cosine	Ny	0.000	
Nozzle Centerline Direction Cosine	Nz	0.000	
Global Force (SUS)	Fx	0.0	Kgf
Global Force (SUS)	Fy	372.9	Kgf
Global Force (SUS)	Fz	0.0	Kgf
Global Moment (SUS)	Mx	0.0	Kg-m.
Global Moment (SUS)	My	0.0	Kg-m.
Global Moment (SUS)	Mz	16.5	Kg-m.

Internal Pressure (SUS)	P	30.50	bars
Include Pressure Thrust		No	

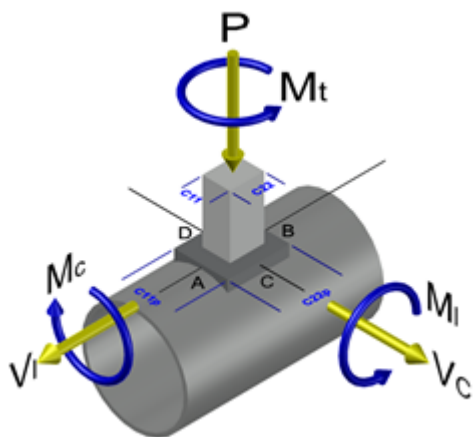
Global Force (OCC)	Fx	45.2	Kgf
Global Force (OCC)	Fy	86.7	Kgf
Global Force (OCC)	Fz	0.0	Kgf
Global Moment (OCC)	Mx	0.0	Kg-m.
Global Moment (OCC)	My	0.0	Kg-m.
Global Moment (OCC)	Mz	49.0	Kg-m.

Occasional Internal Pressure (OCC)	Pvar	0.00	bars
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Use Interactive Control	No		
WRC107 Version	Version	March	1979

Include Pressure Stress Indices per Div. 2	No		
Compute Pressure Stress per WRC-368	No		
Local Loads applied at end of Nozzle/Attachment	No		

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

WRC 107 Stress Calculation for SUSained loads:

Radial Load	P	0.0	Kgf
Circumferential Shear	VC	-0.0	Kgf
Longitudinal Shear	VL	372.9	Kgf
Circumferential Moment	MC	-0.0	Kg-m.
Longitudinal Moment	ML	-16.5	Kg-m.
Torsional Moment	MT	-0.0	Kg-m.

Dimensionless Parameters used : Gamma = 17.00

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
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N(PHI) / (P/Rm)	0.207	4C	2.731	(A,B)
N(PHI) / (P/Rm)	0.207	3C	2.149	(C,D)
M(PHI) / (P)	0.144	2C1	0.091	(A,B)
M(PHI) / (P)	0.144	1C	0.123	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.139	3A	0.482	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.152	1A	0.096	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.176	3B	1.840	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.165	1B	0.045	(A,B,C,D)
N(x) / (P/Rm)	0.178	3C	2.369	(A,B)
N(x) / (P/Rm)	0.178	4C	2.846	(C,D)
M(x) / (P)	0.183	1C1	0.104	(A,B)
M(x) / (P)	0.183	2C	0.070	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.139	4A	0.700	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.191	2A	0.050	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.176	4B	0.561	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.198	2B	0.062	(A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		0.2	0.2	-0.2	-0.2	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		0.6	-0.6	-0.6	0.6	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		0.7	-0.4	-0.7	0.4	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		0.1	0.1	-0.1	-0.1	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		0.6	-0.6	-0.6	0.6	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		0.7	-0.6	-0.7	0.6	0.0	0.0	0.0	0.0
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Str. Int.		0.7	0.6	0.7	0.6	0.5	0.5	0.5	0.5

Dimensionless Parameters used : $\Gamma = 35.33$

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.239	4C	4.598	(A,B)
N(PHI) / (P/Rm)	0.239	3C	2.686	(C,D)
M(PHI) / (P)	0.178	2C1	0.046	(A,B)
M(PHI) / (P)	0.178	1C	0.076	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.175	3A	1.404	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.191	1A	0.081	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.207	3B	3.569	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.199	1B	0.027	(A,B,C,D)



N(x)	/ (P/Rm)	0.211	3C	3.114	(A,B)
N(x)	/ (P/Rm)	0.211	4C	4.893	(C,D)
M(x)	/ (P)	0.216	1C1	0.064	(A,B)
M(x)	/ (P)	0.216	2C !	0.033	(C,D)
N(x)	/ (MC/(Rm**2 * Beta))	0.175	4A	2.442	(A,B,C,D)
M(x)	/ (MC/(Rm * Beta))	0.228	2A	0.035	(A,B,C,D)
N(x)	/ (ML/(Rm**2 * Beta))	0.207	4B	1.446	(A,B,C,D)
M(x)	/ (ML/(Rm * Beta))	0.226	2B	0.035	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		0.7	0.7	-0.7	-0.7	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		1.2	-1.2	-1.2	1.2	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		1.9	-0.6	-1.9	0.6	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		0.3	0.3	-0.3	-0.3	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		1.4	-1.4	-1.4	1.4	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		1.8	-1.1	-1.8	1.1	0.0	0.0	0.0	0.0
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Str. Int.		1.9	1.1	1.9	1.1	1.0	1.0	1.0	1.0

WRC 107 Stress Calculation for OCCasional loads:

Radial Load	P	45.2	Kgf
Circumferential Shear	VC	-0.0	Kgf
Longitudinal Shear	VL	86.7	Kgf
Circumferential Moment	MC	-0.0	Kg-m.
Longitudinal Moment	ML	-49.0	Kg-m.
Torsional Moment	MT	-0.0	Kg-m.

Dimensionless Parameters used : Gamma = 17.00

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1
Circ. Bend. P		-0.3	0.3	-0.3	0.3	-0.4	0.4	-0.4	0.4
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		0.5	0.5	-0.5	-0.5	-0.0	-0.0	-0.0	-0.0



Circ. Bend. ML	1.6	-1.6	-1.6	1.6	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.	1.8	-0.9	-2.5	1.3	-0.4	0.3	-0.4	0.3
Long. Memb. P	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1
Long. Bend. P	-0.3	0.3	-0.3	0.3	-0.2	0.2	-0.2	0.2
Long. Memb. MC	-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC	-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML	0.2	0.2	-0.2	-0.2	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML	1.9	-1.9	-1.9	1.9	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.	1.8	-1.4	-2.5	1.9	-0.3	0.1	-0.3	0.1
Shear VC	-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Shear MT	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Str. Int.	1.8	1.4	2.5	1.9	0.4	0.3	0.4	0.3

Dimensionless Parameters used : Gamma = 35.33

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.3	-0.3	-0.3	-0.3	-0.2	-0.2	-0.2	-0.2
Circ. Bend. P		-0.6	0.6	-0.6	0.6	-1.0	1.0	-1.0	1.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		1.9	1.9	-1.9	-1.9	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		3.7	-3.7	-3.7	3.7	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		4.8	-1.5	-6.5	2.1	-1.1	0.8	-1.1	0.8
Long. Memb. P		-0.2	-0.2	-0.2	-0.2	-0.3	-0.3	-0.3	-0.3
Long. Bend. P		-0.8	0.8	-0.8	0.8	-0.4	0.4	-0.4	0.4
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		1.0	1.0	-1.0	-1.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		4.3	-4.3	-4.3	4.3	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		4.3	-2.6	-6.3	3.9	-0.7	0.1	-0.7	0.1
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Str. Int.		4.8	2.6	6.5	3.9	1.2	0.8	1.2	0.8

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		48.8	51.9	48.8	51.9	48.8	51.9	48.8	51.9
Circ. Pm (OCC)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pm(TOTAL)		48.8	51.9	48.8	51.9	48.8	51.9	48.8	51.9



Circ. P1 (SUS)	0.2	0.2	-0.2	-0.2	0.0	0.0	-0.0	-0.0
Circ. P1 (OCC)	0.5	0.5	-0.6	-0.6	-0.1	-0.1	-0.1	-0.1
Circ. P1(TOTAL)	0.6	0.6	-0.8	-0.8	-0.1	-0.1	-0.1	-0.1
Circ. Q (SUS)	0.6	-0.6	-0.6	0.6	0.0	0.0	-0.0	0.0
Circ. Q (OCC)	1.4	-1.4	-1.9	1.9	-0.4	0.4	-0.4	0.4
Circ. Q (TOTAL)	1.9	-1.9	-2.4	2.4	-0.4	0.4	-0.4	0.4
Long. Pm (SUS)	24.4	24.4	24.4	24.4	24.4	24.4	24.4	24.4
Long. Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pm(TOTAL)	24.4	24.4	24.4	24.4	24.4	24.4	24.4	24.4
Long. P1 (SUS)	0.1	0.1	-0.1	-0.1	0.0	0.0	-0.0	-0.0
Long. P1 (OCC)	0.2	0.2	-0.3	-0.3	-0.1	-0.1	-0.1	-0.1
Long. P1(TOTAL)	0.3	0.3	-0.4	-0.4	-0.1	-0.1	-0.1	-0.1
Long. Q (SUS)	0.6	-0.6	-0.6	0.6	0.0	0.0	-0.0	0.0
Long. Q (OCC)	1.6	-1.6	-2.2	2.2	-0.2	0.2	-0.2	0.2
Long. Q (TOTAL)	2.2	-2.2	-2.8	2.8	-0.2	0.2	-0.2	0.2
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm(TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)	0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Shear P1 (OCC)	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Shear P1(TOTAL)	0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Shear Q (SUS)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (OCC)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	48.8	51.9	48.8	51.9	48.8	51.9	48.8	51.9
Pm (SUS+OCC)	48.8	51.9	48.8	51.9	48.8	51.9	48.8	51.9
Pm+P1 (SUS)	49.0	52.1	48.7	51.7	48.9	51.9	48.9	51.9
Pm+P1 (SUS+OCC)	49.5	52.5	48.1	51.1	48.8	51.8	48.8	51.8
Pm+P1+Q (Total)	51.4	50.6	45.6	53.6	48.4	52.2	48.4	52.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	51.90	138.00	Passed
Pm (SUS+OCC)	51.90	165.60	Passed
Pm+P1 (SUS)	52.08	207.00	Passed
Pm+P1 (SUS+OCC)	52.54	248.40	Passed
Pm+P1+Q (TOTAL)	53.56	485.22	Passed

The Pm+P1+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+P1+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:**Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		104.7	107.8	104.7	107.8	104.7	107.8	104.7	107.8
Circ. Pm (OCC)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pm(TOTAL)		104.7	107.8	104.7	107.8	104.7	107.8	104.7	107.8
Circ. P1 (SUS)		0.7	0.7	-0.7	-0.7	0.0	0.0	-0.0	-0.0
Circ. P1 (OCC)		1.7	1.7	-2.2	-2.2	-0.2	-0.2	-0.2	-0.2
Circ. P1(TOTAL)		2.3	2.3	-2.9	-2.9	-0.2	-0.2	-0.2	-0.2



Circ. Q (SUS)	1.2	-1.2	-1.2	1.2	0.0	0.0	-0.0	0.0
Circ. Q (OCC)	3.1	-3.1	-4.3	4.3	-1.0	1.0	-1.0	1.0
Circ. Q (TOTAL)	4.4	-4.4	-5.5	5.5	-1.0	1.0	-1.0	1.0
Long. Pm (SUS)	52.4	52.4	52.4	52.4	52.4	52.4	52.4	52.4
Long. Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pm(TOTAL)	52.4	52.4	52.4	52.4	52.4	52.4	52.4	52.4
Long. P1 (SUS)	0.3	0.3	-0.3	-0.3	0.0	0.0	-0.0	-0.0
Long. P1 (OCC)	0.8	0.8	-1.2	-1.2	-0.3	-0.3	-0.3	-0.3
Long. P1(TOTAL)	1.2	1.2	-1.6	-1.6	-0.3	-0.3	-0.3	-0.3
Long. Q (SUS)	1.4	-1.4	-1.4	1.4	0.0	0.0	-0.0	0.0
Long. Q (OCC)	3.5	-3.5	-5.1	5.1	-0.4	0.4	-0.4	0.4
Long. Q (TOTAL)	4.9	-4.9	-6.6	6.6	-0.4	0.4	-0.4	0.4
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm(TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)	0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Shear P1 (OCC)	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Shear P1(TOTAL)	0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Shear Q (SUS)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (OCC)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	104.7	107.8	104.7	107.8	104.7	107.8	104.7	107.8
Pm (SUS+OCC)	104.7	107.8	104.7	107.8	104.7	107.8	104.7	107.8
Pm+P1 (SUS)	105.4	108.4	104.1	107.1	104.7	107.8	104.7	107.8
Pm+P1 (SUS+OCC)	107.1	110.1	101.9	104.9	104.6	107.6	104.6	107.6
Pm+P1+Q (Total)	111.4	105.7	96.4	110.5	103.6	108.6	103.6	108.6

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	107.79	138.00	Passed
Pm (SUS+OCC)	107.79	165.60	Passed
Pm+P1 (SUS)	108.44	207.00	Passed
Pm+P1 (SUS+OCC)	110.10	248.40	Passed
Pm+P1+Q (TOTAL)	111.41	485.22	Passed

The Pm+P1+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+P1+Q allowable was based on a temperature range cycling from ambient to design temperature.

Bolting Size Requirement for Leg Baseplates :

Baseplate Material	SA-516 70
Baseplate Allowable Stress	SBA 137.90 N./mm ²
Baseplate Length	B 204.0000 mm.
Baseplate Width	D 204.0000 mm.
Baseplate Thickness	BTHK 18.0000 mm.
Leg Dimension Along Baseplate Length	d 79.9998 mm.
Leg Dimension Along Baseplate Width	b 79.9998 mm.
Bolt Circle Diameter	BCD 982.0000 mm.
Bolt Material	SA-307 B
Bolt Allowable Tensile Stress	STBA 138.00 N./mm ²
Bolt Allowable Shear Stress	SBShear 82.80 N./mm ²
Anchor Bolt Nominal Diameter	BOD 24.0000 mm.
Number of Anchor Bolts in Tension per Leg	NB 0
Total Number of Anchors Bolt per Leg	NBT 1
Bolt Corrosion Allowance	bca 0.0000 mm.
Ultimate 28-day Concrete Strength	FCPRIME 50.000 N./mm ²

**Leg Baseplate and Bolting Analysis, including Moments**

Shear Stress in a Single Bolt [tb]:

= Shear Force/(Bolt Area • Number of Bolts)

= 114/(2.4 • 4)

= 1.2 N./mm². Must be less than 82.8 N./mm².

Plate thickness required (5/8"; 16mm) per D. Moss

Since there is only 1 bolt, the standard computation for this geometry is not possible. A reasonable bolt size should be selected. Per Megyesy, use a 1 inch (25.4 mm) bolt.

Review notes about nozzle loadings and supports in the Warnings report.

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RESULTS FOR LEGS : HydroTest Case Description: LEGS

Legs attached to: Shell

Section Properties : Single Angle ISA80X80X10

India ISI Structural Steel Data

Overall Leg Length		2300.000	mm.
Effective Leg Length	Leglen	2000.000	mm.
Distance Leg Up Side of Vessel		250.000	mm.
Number of Legs	Nleg	4	
Cross Sectional Area for ISA80X80X10	Aleg	15.050	cm ²
Section Inertia (strong axis)		87.700	cm ⁴
Section Inertia (weak axis)		87.700	cm ⁴
Section Modulus (strong axis)		15500.063	mm. ³
Section Modulus (weak axis)		15500.063	mm. ³
Radius of Gyration (strong axis)		24.100	mm.
Radius of Gyration (weak axis)		24.100	mm.

Leg Orientation - Diagonal

Overturning Moment at top of Legs		39.9	Kg-m.
Total Weight Load at top of Legs	W	3320.9	Kgf
Total Shear force at top of Legs		57.0	Kgf
Additional force in Leg due to Bracing	Fadd	0.0	Kgf
Occasional Load Factor	Occfac	1.333	
Effective Leg End Condition Factor	k	1.000	

Note: The Legs are Not Cross Braced

The Leg Shear Force includes Wind, Seismic, and Applied Load Effects.

Pad Width along Circumference	C11P	150.000	mm.
Pad Length along Vessel Axis	C22P	250.000	mm.
Pad Thickness	Tpad	16.000	mm.

Maximum Shear at top of one Leg [Vleg]:
 $= (\max(\text{Wind, Seismic, Blast}) + \text{applied forces}) (I_{\max} / I_{\text{tot}})$
 $= (57) (141/356)$
 $= 22.60 \text{ Kgf}$

Axial Compression, Leg furthest from the Neutral Axis [Sma]:
 $= (W/N_{\text{leg}} + (M_{\text{leg}} / (N_{\text{legm}} \cdot R_n))) / A_{\text{leg}}$
 $= (32567/4 + (390983 / (2 \cdot 560))) / 1505$
 $= 5.64 \text{ N./mm}^2$

Axial Compression, Leg closest to the Neutral Axis [Sva]:
 $= (W / N_{\text{leg}}) / A_{\text{leg}}$
 $= (3321/4) / 15$
 $= 5.41 \text{ N./mm}^2$

Computing Principal Axis and Inertias for Angle:

Leg lengths and thickness:	80.000	80.000	10.000
Distance to geometric centroid:	23.400	23.400	
Arm about YY:	18.400	21.600	
Arm about ZZ:	16.600	18.400	
Leg areas:	8.0000	7.0000	
Geometric inertia components YY:	27.751	61.242	
Geometric inertia components ZZ:	64.711	24.282	
Geometric inertias Iy & Iz:	88.993	88.993	
Product of inertia:	52.256		
Mohrs Radius:	52.256		
Average Inertia:	88.993		

QFACT = 1.0000 FBZ = 0.16383
 Principal Axis Inertias (Z&W) = 36.738 141.25



Angle to Principal Axis = 45.000
 Distances to extreme fibers CW & CZ = 56.568 23.476
 FOB from Eq 5-5 = 0.97392
 Bending allowables Fby & Fbz = 0.16383 0.16383

Shear Center Coordinates Wo & Zo: 25.095 0.0000

Values for Elastic Flexural-Torsional Buckling Stress:

E, G, J, R0²: 29500. 11346. 0.12053 2.8092
 AREA, LENGTH, Kw, Kz: 2.3328 78.740 1.0000 1.0000
 H, Few, Fez, Fej: 0.65252 68.314 17.768 208.68
 Fe computed from C4-1: 59.926

Initial (Kl/r)max, & (Kl/r)equiv = 128.01 69.704
 Final (Kl/r)max, & Cc = 128.01 127.18
 Fa based on Eq 4-2 = 0.63918E-01

Leg Bending and Axial Stresses vs. Allowables: (N./mm²)

	Actual	Allowable
Weak Axis Bending :	20.03	218.38
Strong Axis Bending :	12.55	218.38
Axial Compression :	5.64	85.20

UNITY CHECKS ARE: H1-1 0.000
 H1-2 0.000
 H1-3 0.215

AISC Unity Check : 0.215, should be <= to 1 for Code compliance, [Passed]

WRC 107 Stress Analysis for Leg to Shell Junction, Test Condition

Rectangular Attachment Parameter C11 113.137 mm.
 Rectangular Attachment Parameter C22 230.950 mm.

Input Echo, WRC107/537 Item 1, Description: LEGS

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Cylindrical	
Internal Corrosion Allowance	Cas	1.6000	mm.
Vessel Diameter	Dv	1000.000	mm.
Vessel Thickness	Tv	16.000	mm.
Design Temperature	T1	82.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	234.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	234.00	N./mm ²
Attachment Type	Type	Rectangular	
Parameter C11	C11	113.14	mm.
Parameter C22	C22	230.95	mm.
Thickness of Reinforcing Pad	Tpad	16.000	mm.
Pad Parameter C11p	C11p	150.000	mm.
Pad Parameter C22p	C22p	250.000	mm.
Design Internal Pressure	Dp	30.500	bars
Include Pressure Thrust		No	
Vessel Centerline Direction Cosine	Vx	0.000	
Vessel Centerline Direction Cosine	Vy	1.000	
Vessel Centerline Direction Cosine	Vz	0.000	
Nozzle Centerline Direction Cosine	Nx	1.000	
Nozzle Centerline Direction Cosine	Ny	0.000	
Nozzle Centerline Direction Cosine	Nz	0.000	



Global Force (SUS)	Fx	0.0	Kgf
Global Force (SUS)	Fy	830.2	Kgf
Global Force (SUS)	Fz	0.0	Kgf
Global Moment (SUS)	Mx	0.0	Kg-m.
Global Moment (SUS)	My	0.0	Kg-m.
Global Moment (SUS)	Mz	36.7	Kg-m.

Internal Pressure (SUS)	P	30.50	bars
Include Pressure Thrust		No	

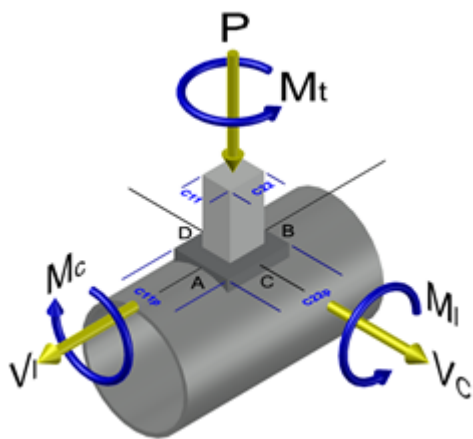
Global Force (OCC)	Fx	22.6	Kgf
Global Force (OCC)	Fy	35.6	Kgf
Global Force (OCC)	Fz	0.0	Kgf
Global Moment (OCC)	Mx	0.0	Kg-m.
Global Moment (OCC)	My	0.0	Kg-m.
Global Moment (OCC)	Mz	24.2	Kg-m.

Occasional Internal Pressure (OCC)	Pvar	9.39	bars
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Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

WRC 107 Stress Calculation for SUSstained loads:

Radial Load	P	0.0	Kgf
Circumferential Shear	VC	-0.0	Kgf
Longitudinal Shear	VL	830.2	Kgf
Circumferential Moment	MC	-0.0	Kg-m.
Longitudinal Moment	ML	-36.7	Kg-m.
Torsional Moment	MT	-0.0	Kg-m.

Dimensionless Parameters used : Gamma = 17.00

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.207	4C	2.731	(A,B)
N(PHI) / (P/Rm)	0.207	3C	2.149	(C,D)



M(PHI) / (P)	0.144	2C1	0.091	(A,B)
M(PHI) / (P)	0.144	1C	0.123	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.139	3A	0.482	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.152	1A	0.096	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.176	3B	1.840	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.165	1B	0.045	(A,B,C,D)
N(x) / (P/Rm)	0.178	3C	2.369	(A,B)
N(x) / (P/Rm)	0.178	4C	2.846	(C,D)
M(x) / (P)	0.183	1C1	0.104	(A,B)
M(x) / (P)	0.183	2C	0.070	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.139	4A	0.700	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.191	2A	0.050	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.176	4B	0.561	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.198	2B	0.062	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		0.4	0.4	-0.4	-0.4	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		1.2	-1.2	-1.2	1.2	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		1.6	-0.8	-1.6	0.8	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		0.2	0.2	-0.2	-0.2	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		1.4	-1.4	-1.4	1.4	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		1.6	-1.2	-1.6	1.2	0.0	0.0	0.0	0.0
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Str. Int.		1.6	1.2	1.6	1.2	1.2	1.2	1.2	1.2

Dimensionless Parameters used : Gamma = 35.33

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.239	4C	4.598	(A,B)
N(PHI) / (P/Rm)	0.239	3C	2.686	(C,D)
M(PHI) / (P)	0.178	2C1	0.046	(A,B)
M(PHI) / (P)	0.178	1C	0.076	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.175	3A	1.404	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.191	1A	0.081	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.207	3B	3.569	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.199	1B	0.027	(A,B,C,D)
N(x) / (P/Rm)	0.211	3C	3.114	(A,B)
N(x) / (P/Rm)	0.211	4C	4.893	(C,D)
M(x) / (P)	0.216	1C1	0.064	(A,B)

M(x) / (P)	0.216	2C !	0.033	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.175	4A	2.442	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.228	2A	0.035	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.207	4B	1.446	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.226	2B	0.035	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		1.4	1.4	-1.4	-1.4	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		2.8	-2.8	-2.8	2.8	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		4.2	-1.3	-4.2	1.3	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		0.8	0.8	-0.8	-0.8	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		3.2	-3.2	-3.2	3.2	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		4.0	-2.4	-4.0	2.4	0.0	0.0	0.0	0.0
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-1.1	-1.1	1.1	1.1
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-1.1	-1.1	1.1	1.1
Str. Int.		4.2	2.4	4.2	2.4	2.3	2.3	2.3	2.3

WRC 107 Stress Calculation for OCCasional loads:

Radial Load	P	22.6	Kgf
Circumferential Shear	VC	-0.0	Kgf
Longitudinal Shear	VL	35.6	Kgf
Circumferential Moment	MC	-0.0	Kg-m.
Longitudinal Moment	ML	-24.2	Kg-m.
Torsional Moment	MT	-0.0	Kg-m.

Dimensionless Parameters used : Gamma = 17.00

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.1	0.1	-0.1	0.1	-0.2	0.2	-0.2	0.2
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		0.3	0.3	-0.3	-0.3	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		0.8	-0.8	-0.8	0.8	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		0.9	-0.5	-1.2	0.6	-0.2	0.1	-0.2	0.1



Long. Memb. P	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P	-0.1	0.1	-0.1	0.1	-0.1	0.1	-0.1	0.1
Long. Memb. MC	-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC	-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	0.0
Long. Memb. ML	0.1	0.1	-0.1	-0.1	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML	0.9	-0.9	-0.9	0.9	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.	0.9	-0.7	-1.2	0.9	-0.1	0.1	-0.1	0.1
Shear VC	-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Shear MT	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Str. Int.	0.9	0.7	1.2	0.9	0.2	0.2	0.2	0.2

Dimensionless Parameters used : Gamma = 35.33

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1
Circ. Bend. P		-0.3	0.3	-0.3	0.3	-0.5	0.5	-0.5	0.5
Circ. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Circ. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Circ. Memb. ML		1.0	1.0	-1.0	-1.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. ML		1.8	-1.8	-1.8	1.8	-0.0	-0.0	-0.0	-0.0
Tot. Circ. Str.		2.3	-0.7	-3.2	1.0	-0.6	0.4	-0.6	0.4
Long. Memb. P		-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1
Long. Bend. P		-0.4	0.4	-0.4	0.4	-0.2	0.2	-0.2	0.2
Long. Memb. MC		-0.0	-0.0	-0.0	-0.0	0.0	0.0	-0.0	-0.0
Long. Bend. MC		-0.0	-0.0	-0.0	-0.0	0.0	-0.0	-0.0	0.0
Long. Memb. ML		0.5	0.5	-0.5	-0.5	-0.0	-0.0	-0.0	-0.0
Long. Bend. ML		2.1	-2.1	-2.1	2.1	-0.0	-0.0	-0.0	-0.0
Tot. Long. Str.		2.1	-1.3	-3.1	1.9	-0.4	0.1	-0.4	0.1
Shear VC		-0.0	-0.0	0.0	0.0	-0.0	-0.0	-0.0	-0.0
Shear VL		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Shear MT		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Tot. Shear		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Str. Int.		2.3	1.3	3.2	1.9	0.6	0.4	0.6	0.4

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		48.8	51.9	48.8	51.9	48.8	51.9	48.8	51.9
Circ. Pm (OCC)		15.0	16.0	15.0	16.0	15.0	16.0	15.0	16.0
Circ. Pm(TOTAL)		63.9	67.9	63.9	67.9	63.9	67.9	63.9	67.9
Circ. Pl (SUS)		0.4	0.4	-0.4	-0.4	0.0	0.0	-0.0	-0.0
Circ. Pl (OCC)		0.2	0.2	-0.3	-0.3	-0.0	-0.0	-0.0	-0.0
Circ. Pl(TOTAL)		0.6	0.6	-0.7	-0.7	-0.0	-0.0	-0.0	-0.0



Circ. Q (SUS)	1.2	-1.2	-1.2	1.2	0.0	0.0	-0.0	0.0
Circ. Q (OCC)	0.7	-0.7	-0.9	0.9	-0.2	0.2	-0.2	0.2
Circ. Q (TOTAL)	1.9	-1.9	-2.2	2.2	-0.2	0.2	-0.2	0.2
Long. Pm (SUS)	24.4	24.4	24.4	24.4	24.4	24.4	24.4	24.4
Long. Pm (OCC)	7.5	7.5	7.5	7.5	7.5	7.5	7.5	7.5
Long. Pm(TOTAL)	31.9	31.9	31.9	31.9	31.9	31.9	31.9	31.9
Long. P1 (SUS)	0.2	0.2	-0.2	-0.2	0.0	0.0	-0.0	-0.0
Long. P1 (OCC)	0.1	0.1	-0.2	-0.2	-0.0	-0.0	-0.0	-0.0
Long. P1(TOTAL)	0.3	0.3	-0.3	-0.3	-0.0	-0.0	-0.0	-0.0
Long. Q (SUS)	1.4	-1.4	-1.4	1.4	0.0	0.0	-0.0	0.0
Long. Q (OCC)	0.8	-0.8	-1.1	1.1	-0.1	0.1	-0.1	0.1
Long. Q (TOTAL)	2.2	-2.2	-2.5	2.5	-0.1	0.1	-0.1	0.1
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm(TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)	0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Shear P1 (OCC)	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Shear P1(TOTAL)	0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Shear Q (SUS)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (OCC)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	48.8	51.9	48.8	51.9	48.8	51.9	48.8	51.9
Pm (SUS+OCC)	63.9	67.9	63.9	67.9	63.9	67.9	63.9	67.9
Pm+P1 (SUS)	49.2	52.3	48.4	51.5	48.9	51.9	48.9	51.9
Pm+P1 (SUS+OCC)	64.5	68.5	63.2	67.2	63.9	67.9	63.9	67.9
Pm+P1+Q (Total)	66.4	66.6	61.0	69.3	63.7	68.0	63.7	68.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	51.90	234.00	Passed
Pm (SUS+OCC)	67.88	280.79	Passed
Pm+P1 (SUS)	52.30	350.99	Passed
Pm+P1 (SUS+OCC)	68.50	421.19	Passed
Pm+P1+Q (TOTAL)	69.34	701.99	Passed

The Pm+P1+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+P1+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:**Vessel Stress Summation at Reinforcing Pad Edge (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		104.7	107.8	104.7	107.8	104.7	107.8	104.7	107.8
Circ. Pm (OCC)		32.2	33.2	32.2	33.2	32.2	33.2	32.2	33.2
Circ. Pm(TOTAL)		137.0	141.0	137.0	141.0	137.0	141.0	137.0	141.0
Circ. P1 (SUS)		1.4	1.4	-1.4	-1.4	0.0	0.0	-0.0	-0.0
Circ. P1 (OCC)		0.8	0.8	-1.1	-1.1	-0.1	-0.1	-0.1	-0.1
Circ. P1(TOTAL)		2.3	2.3	-2.5	-2.5	-0.1	-0.1	-0.1	-0.1
Circ. Q (SUS)		2.8	-2.8	-2.8	2.8	0.0	0.0	-0.0	0.0
Circ. Q (OCC)		1.5	-1.5	-2.1	2.1	-0.5	0.5	-0.5	0.5
Circ. Q (TOTAL)		4.3	-4.3	-4.9	4.9	-0.5	0.5	-0.5	0.5



Long. Pm (SUS)	52.4	52.4	52.4	52.4	52.4	52.4	52.4	52.4
Long. Pm (OCC)	16.1	16.1	16.1	16.1	16.1	16.1	16.1	16.1
Long. Pm(TOTAL)	68.5	68.5	68.5	68.5	68.5	68.5	68.5	68.5
Long. P1 (SUS)	0.8	0.8	-0.8	-0.8	0.0	0.0	-0.0	-0.0
Long. P1 (OCC)	0.4	0.4	-0.6	-0.6	-0.1	-0.1	-0.1	-0.1
Long. P1(TOTAL)	1.2	1.2	-1.4	-1.4	-0.1	-0.1	-0.1	-0.1
Long. Q (SUS)	3.2	-3.2	-3.2	3.2	0.0	0.0	-0.0	0.0
Long. Q (OCC)	1.7	-1.7	-2.5	2.5	-0.2	0.2	-0.2	0.2
Long. Q (TOTAL)	4.9	-4.9	-5.7	5.7	-0.2	0.2	-0.2	0.2
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (OCC)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm(TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear P1 (SUS)	0.0	0.0	0.0	0.0	-1.1	-1.1	1.1	1.1
Shear P1 (OCC)	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Shear P1(TOTAL)	0.0	0.0	0.0	0.0	-1.2	-1.2	1.2	1.2
Shear Q (SUS)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (OCC)	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Shear Q (TOTAL)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm (SUS)	104.7	107.8	104.7	107.8	104.7	107.8	104.7	107.8
Pm (SUS+OCC)	137.0	141.0	137.0	141.0	137.0	141.0	137.0	141.0
Pm+P1 (SUS)	106.2	109.2	103.3	106.3	104.8	107.8	104.8	107.8
Pm+P1 (SUS+OCC)	139.3	143.2	134.5	138.4	136.9	140.9	136.9	140.9
Pm+P1+Q (Total)	143.6	138.9	129.6	143.3	136.4	141.4	136.4	141.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	107.79	234.00	Passed
Pm (SUS+OCC)	140.98	280.79	Passed
Pm+P1 (SUS)	109.24	350.99	Passed
Pm+P1 (SUS+OCC)	143.24	421.19	Passed
Pm+P1+Q (TOTAL)	143.56	701.99	Passed

The Pm+P1+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+P1+Q allowable was based on a temperature range cycling from ambient to design temperature.

Bolting Size Requirement for Leg Baseplates :

Baseplate Material	SA-516 70
Baseplate Allowable Stress	SBA 137.90 N./mm ²
Baseplate Length	B 204.0000 mm.
Baseplate Width	D 204.0000 mm.
Baseplate Thickness	BTHK 18.0000 mm.
Leg Dimension Along Baseplate Length	d 79.9998 mm.
Leg Dimension Along Baseplate Width	b 79.9998 mm.
Bolt Circle Diameter	BCD 982.0000 mm.
Bolt Material	SA-307 B
Bolt Allowable Tensile Stress	STBA 138.00 N./mm ²
Bolt Allowable Shear Stress	SBShear 82.80 N./mm ²
Anchor Bolt Nominal Diameter	BOD 24.0000 mm.
Number of Anchor Bolts in Tension per Leg	NB 0
Total Number of Anchors Bolt per Leg	NBT 1
Bolt Corrosion Allowance	bca 0.0000 mm.
Ultimate 28-day Concrete Strength	FCPRIME 50.000 N./mm ²

Leg Baseplate and Bolting Analysis, including Moments

Shear Stress in a Single Bolt [tb]:



= Shear Force/(Bolt Area • Number of Bolts)
= 114/(2.4 • 4)
= 1.2 N./mm². Must be less than 82.8 N./mm².

Plate thickness required (5/8"; 16mm) per D. Moss

Since there is only 1 bolt, the standard computation for this geometry is not possible. A reasonable bolt size should be selected. Per Megyesy, use a 1 inch (25.4 mm) bolt.

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N2-2"

From: 10

Pressure for Reinforcement Calculations	P	30.500	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	1000.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	16.0000	mm.
Head Internal Corrosion Allowance	c	1.6000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: N2-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.375	in.
Actual Thickness Used in Calculation	0.301	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 30.5 \cdot 1003 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 30.5)$
 $= 9.9701$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]



$$\begin{aligned} &= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)} \\ &= 30.5 \cdot 30.2 / (118 \cdot 1 + 0.4 \cdot 30.5) \\ &= 0.7717 \text{ mm.} \end{aligned}$$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	96.4684	mm.
Parallel to Vessel Wall, opening length	d	48.2342	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	15.1135	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: N2-2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.3717	mm.
Wall Thickness per UG16(b),	tr16b	= 3.1000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 12.6646	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 12.6646	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.1000	mm.
Wall Thickness per table UG-45	tb3	= 5.0200	mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[5.02, max(12.7, 3.1)]
= 5.0200 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(2.37, 5.02)
= 5.0200 mm.

Available Nozzle Neck Thickness = 7.6454 mm. -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code,
(see 319.4.4 and 302.3.5):**

Sustained	:	47.9,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	247.1 N./mm ²	Passed
Occasional	:	5.4,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	30.3,	Allowable	:	82.6 N./mm ²	Passed

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of
the expansion stress allowable.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 7.65, tr = 0.77, c = 1.6 mm., E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.13, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 7.65, tr = 0.77, c = 1.6 mm., E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.13, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:



Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 30.50/102.00 = 0.299

Weld Size Calculations, Description: N2-2"

Intermediate Calc. for nozzle/shell Welds Tmin 6.0454 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	4.2318 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.4985 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 216.4985 mm.

Input Echo, WRC107/537 Item 1, Description: N2-2" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.6000	mm.
Vessel Diameter	Dv	1800.000	mm.
Vessel Thickness	Tv	16.000	mm.
Design Temperature	T1	82.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	1.6000	mm.
Nozzle Diameter	Dn	60.325	mm.
Nozzle Thickness	Tn	7.645	mm.
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm ²
Design Internal Pressure	Dp	30.500	bars
Include Pressure Thrust		No	

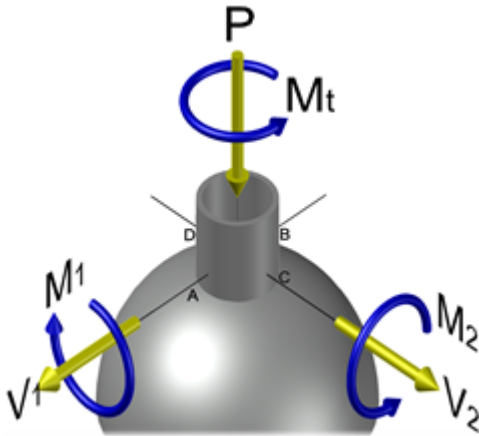
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	382.4	Kgf
Longitudinal Shear (SUS)	(V1) V1	254.9	Kgf
Circumferential Shear (SUS)	(Vc) V2	254.9	Kgf
Circumferential Moment (SUS)	(Mc) M1	35.7	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	35.7	Kg-m.
Torsional Moment (SUS)	Mt	51.0	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979
Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No



WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 60.325 + 2 \cdot 1.65 \cdot \sqrt{(908.8 (16.0 - 1.6))}$$

$$= 437.836 \text{ mm.}$$

WRC 107 Stress Calculation for SUSstained loads:

Radial Load	P	382.4	Kgf
Circumferential Shear	(VC) V2	254.9	Kgf
Longitudinal Shear	(VL) V1	254.9	Kgf
Circumferential Moment	(MC) M1	35.7	Kg-m.
Longitudinal Moment	(ML) M2	35.7	Kg-m.
Torsional Moment	MT	51.0	Kg-m.

Dimensionless Param: U = 0.26 TAU = 5.00 (4.49) RHO = 2.38

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 3	0.11165	(A,B,C,D)
$M(x) / P$	SP 3	0.07600	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 3	0.32076	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 3	0.24281	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 3	0.32076	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 3	0.24281	(A,B,C,D)
$N(y) * T / P$	SP 3	0.25357	(A,B,C,D)
$M(y) / P$	SP 3	0.10127	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 3	0.10100	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 3	0.46546	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 3	0.10100	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 3	0.46546	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	-2.0	-2.0	-2.0	-2.0	-2.0	-2.0	-2.0	-2.0
Rad. Bend.	P	-8.2	8.2	-8.2	8.2	-8.2	8.2	-8.2	8.2
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-4.7	-4.7	4.7	4.7



Rad. Bend. MC	0.0	0.0	0.0	0.0	-21.5	21.5	21.5	-21.5
Rad. Memb. ML	-4.7	-4.7	4.7	4.7	0.0	0.0	0.0	0.0
Rad. Bend. ML	-21.5	21.5	21.5	-21.5	0.0	0.0	0.0	0.0
Tot. Rad. Str.	-36.5	23.0	16.0	-10.5	-36.5	23.0	16.0	-10.5
Tang. Memb. P	-4.6	-4.6	-4.6	-4.6	-4.6	-4.6	-4.6	-4.6
Tang. Bend. P	-11.0	11.0	-11.0	11.0	-11.0	11.0	-11.0	11.0
Tang. Memb. MC	0.0	0.0	0.0	0.0	-1.5	-1.5	1.5	1.5
Tang. Bend. MC	0.0	0.0	0.0	0.0	-41.2	41.2	41.2	-41.2
Tang. Memb. ML	-1.5	-1.5	1.5	1.5	0.0	0.0	0.0	0.0
Tang. Bend. ML	-41.2	41.2	41.2	-41.2	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-58.3	46.1	27.1	-33.3	-58.3	46.1	27.1	-33.3
Shear VC	1.8	1.8	-1.8	-1.8	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-1.8	-1.8	1.8	1.8
Shear MT	6.1	6.1	6.1	6.1	6.1	6.1	6.1	6.1
Tot. Shear	7.9	7.9	4.2	4.2	4.2	4.2	7.9	7.9
Str. Int.	60.8	48.6	28.5	34.1	59.1	46.9	31.2	35.8

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		95.5	95.5	95.5	95.5	95.5	95.5	95.5	95.5
Rad. Pl (SUS)		-6.8	-6.8	2.7	2.7	-6.8	-6.8	2.7	2.7
Rad. Q (SUS)		-29.7	29.7	13.2	-13.2	-29.7	29.7	13.2	-13.2
Long. Pm (SUS)		95.5	95.5	95.5	95.5	95.5	95.5	95.5	95.5
Long. Pl (SUS)		-6.1	-6.1	-3.1	-3.1	-6.1	-6.1	-3.1	-3.1
Long. Q (SUS)		-52.2	52.2	30.2	-30.2	-52.2	52.2	30.2	-30.2
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		1.8	1.8	-1.8	-1.8	-1.8	-1.8	1.8	1.8
Shear Q (SUS)		6.1	6.1	6.1	6.1	6.1	6.1	6.1	6.1
Pm (SUS)		95.5	95.5	95.5	95.5	95.5	95.5	95.5	95.5
Pm+Pl (SUS)		91.0	91.0	98.7	98.7	91.0	91.0	98.7	98.7
Pm+Pl+Q (Total)		61.6	144.1	124.0	85.7	59.8	142.4	126.7	87.5

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	95.50	138.00	Passed
Pm+Pl (SUS)	98.75	207.00	Passed
Pm+Pl+Q (TOTAL)	144.06	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N1-2"

From: 20

Pressure for Reinforcement Calculations	P	30.500	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1000.00	mm.
Shell Finished (Minimum) Thickness	t	16.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1750.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: N1-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.375	in.
Actual Thickness Used in Calculation	0.301	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (Sv \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 30.5 \cdot 502 / (138 \cdot 1 - 0.6 \cdot 30.5)$
 $= 11.2360$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)



$$= 30.5 \cdot 30.2 / (118 \cdot 1 + 0.4 \cdot 30.5)$$
$$= 0.7717 \text{ mm.}$$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	96.4684	mm.
Parallel to Vessel Wall, opening length	d	48.2342	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	15.1135	mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: N1-2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.3717	mm.
Wall Thickness per UG16(b),	tr16b	= 3.1000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 12.8360	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 12.8360	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.1000	mm.
Wall Thickness per table UG-45	tb3	= 5.0200	mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$
$$= \min[5.02, \max(12.8, 3.1)]$$
$$= 5.0200 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$
$$= \max(2.37, 5.02)$$
$$= 5.0200 \text{ mm.}$$

Available Nozzle Neck Thickness = 7.6454 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	47.9,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	247.1 N./mm ²	Passed
Occasional	:	5.4,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	30.3,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 7.65, tr = 0.77, c = 1.6 mm., E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.13, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 7.65, tr = 0.77, c = 1.6 mm., E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.13, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:



Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 30.50/102.00 = 0.299

Weld Size Calculations, Description: N1-2"

Intermediate Calc. for nozzle/shell Welds Tmin 6.0454 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	4.2318 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.9106 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 216.9106 mm.

Input Echo, WRC297 Item 1, Description: N1-2" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.6000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1000.000	mm.
Shell Thickness	16.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.6000	mm.
Nozzle Diameter	60.325	mm.
Nozzle Thickness	7.6454	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	30.50	bars
Radial Load	P	382.42	Kgf
Circumferential Shear	Vc	254.95	Kgf
Longitudinal Shear	Vl	254.95	Kgf
Circumferential Moment	Mc	35.68	Kg-m.
Longitudinal Moment	Ml	35.68	Kg-m.
Torsional Moment	Mt	50.97	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tv, (3.770) is < 5
 check the Limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (7.890) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:



Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 60.325 + 2 \cdot 1.65 \cdot \sqrt{508.8(16.0 - 1.6)}$$

$$= 342.793 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1017.600 mm.
 Nozzle Outside Diameter (d) = 60.325 mm.
 Vessel Thickness used (T) = 14.400 mm.
 Nozzle Thickness used (t) = 6.045 mm.
 T / t = 2.382
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.498$$

Nr/P = 0.199
 Mr/P = 0.181
 M0/P = 0.134
 N0/P = 0.173
 MrD/Mc = 0.212
 NrDL/Mc = 0.193
 M0d/Mc = 0.189
 N0DL/Mc = -0.101
 MrD/M1 = 0.197
 NrDL/M1 = 0.186
 M0D/M1 = 0.176
 N0DL/M1 = -0.034

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-3	-3	-3	-3
Outplane Bending (P)	-14	14	-14	14
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-29	29	29	-29
Normal Pressure Stress	104	107	104	107
Outplane Stress Summary	58	149	115	88

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-3	-3	-3	-3
Inplane Bending (P)	-19	19	-19	19
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-33	33	33	-33
Inplane Pressure Stress	52	52	52	52
Inplane Stress Summary	-9	96	67	40

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	6	6	6	6

Shear Stress Summary 7 7 4 4

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE
 (Stress Intensities)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress	69	150	116	89
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to
 circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	-3	-3	-3	-3
Outplane Bending (P)	-14	14	-14	14
Outplane Membrane (Mc)	2	2	-2	-2
Outplane Bending (Mc)	-31	31	31	-31
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	52	52	52	52

Outplane Stress Summary 5 98 63 29

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stresses parallel to
 circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	-3	-3	-3	-3
Inplane Bending (P)	-19	19	-19	19
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-35	35	35	-35
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	104	107	104	107

Inplane Stress Summary 40 153 122 93

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Shear stress normal to
 circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	6	6	6	6

Shear Stress Summary 4 4 7 7

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stress Intensities)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	41	154	123	94
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE
 (Stresses in the
 hoop direction)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	-3	-3	-3	-3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	12	16	12	16



Hoop Stress Summary	10	14	8	12
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	-3	-3	-3	-3
Axial Bending (P)	-49	49	-49	49
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-27	-27	27	27
Axial Bending (ML)	-99	99	99	-99
Axial Pressure Stress	6	6	6	6

Axial Stress Summary	-173	124	79	-18
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	4	4	-4	-4
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	14	14	14	14

Shear Stress Summary	18	18	10	10
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	188	127	80	37

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-3	-3	-3	-3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	2	2	-2	-2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	12	16	12	16

Hoop Stress Summary	12	16	7	10
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	-3	-3	-3	-3
Axial Bending (P)	-49	49	-49	49
Axial Membrane (Mc)	-27	-27	27	27
Axial Bending (Mc)	-109	109	109	-109
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6

Axial Stress Summary	-184	134	90	-29
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0



Shear due to (V1)	-4	-4	4	4
Shear due to Torsion	14	14	14	14
Shear Stress Summary	10	10	18	18

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	197	135	94	55

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	104	107	104	107	104	107	104	107
Circ. Pl (SUS)	-3	-3	-3	-3	-8	-8	2	2
Circ. Q (SUS)	-43	43	15	-15	-54	54	16	-16
Long. Pm (SUS)	52	52	52	52	52	52	52	52
Long. Pl (SUS)	-8	-8	2	2	-1	-1	-5	-5
Long. Q (SUS)	-52	52	14	-14	-45	45	17	-17
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	104.0	107.0	104.0	107.0	104.0	107.0	104.0	107.0
Pm+Pl (SUS)	101.0	104.0	101.0	104.0	96.0	99.0	106.0	109.0
Pm+Pl+Q (Total)	67.5	147.9	116.5	89.5	42.7	153.4	122.8	93.8

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	107.00	138.00	Passed
Pm+Pl (SUS)	109.02	207.00	Passed
Pm+Pl+Q (TOTAL)	153.44	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	12	16	12	16	12	16	12	16
Circ. Pl (SUS)	-3	-3	-3	-3	-1	-1	-5	-5
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	-30	-30	24	24	-30	-30	24	24
Long. Q (SUS)	-148	148	50	-50	-158	158	60	-60
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	4	4	-4	-4	-4	-4	4	4
Shear Q (SUS)	14	14	14	14	14	14	14	14
Pm (SUS)	12.0	16.0	12.0	16.0	12.0	16.0	12.0	16.0



Pm+Pl (SUS)	34.0	37.9	30.7	30.9	35.9	39.8	30.7	30.8
Pm+Pl+Q (Total)	184.5	126.8	81.4	38.6	194.0	134.8	93.7	54.6

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	16.00	118.00	Passed
Pm+Pl (SUS)	39.81	177.00	Passed
Pm+Pl+Q (TOTAL)	194.03	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: M1-6"

From: 20

Pressure for Reinforcement Calculations	P	30.500	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1000.00	mm.
Shell Finished (Minimum) Thickness	t	16.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		800.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	268.2750	mm.
Thickness of Pad	te	16.0000	mm.
Weld leg size between Pad and Shell	Wp	12.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm.
Reinforcing Pad Width		50.0000	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: M1-6"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.625	in.
Actual Thickness Used in Calculation	0.628	in.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 30.5 \cdot 502 / (138 \cdot 1 - 0.6 \cdot 30.5)$
 $= 11.2360 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 30.5 \cdot 84.1 / (118 \cdot 1 + 0.4 \cdot 30.5)$
 $= 2.1526 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 279.1198 mm.
 Parallel to Vessel Wall, opening length d 139.5599 mm.
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 36.0000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	16.149	NA	NA
Area in Shell	A1	4.284	NA	NA
Area in Nozzle Wall	A2	7.514	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		1.862	NA	NA
Area in Element	A5	16.000	NA	NA
TOTAL AREA AVAILABLE	Atot	29.661	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 183.8247 16.0000 mm.
 Based on given Pad Diameter: 268.2750 2.4879 mm.
 Based on Shell or Nozzle Thickness: 183.8660 15.9576 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (140 \cdot 11.2 \cdot 1 + 2 \cdot 14.4 \cdot 11.2 \cdot 1 \cdot (1 - 0.86))$
 $= 16.149 \text{ cm}^2$

**Reinforcement Areas per Figure UG-37.1**

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 140(1 \cdot 14.4 - 1 \cdot 11.2) - 2 \cdot 14.4$$

$$(1 \cdot 14.4 - 1 \cdot 11.2) \cdot (1 - 0.86)$$

$$= 4.284 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 36) \cdot (14.4 - 2.15) \cdot 0.86$$

$$= 7.514 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (Wo^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi - can/0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Trapfr4$$

$$= (1) \cdot 0.86 + (0) \cdot 0.86 + 101^2 \cdot 1$$

$$= 1.862 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$$

$$= (268 - 168) \cdot 16 \cdot 1$$

$$= 16.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:Wall Thickness for Internal/External pressures $t_a = 3.7526 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 3.7526 \text{ mm.}$$

Available Nozzle Neck Thickness = 15.9576 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)**

Govrn. thk, tg = 16, tr = 2.15, c = 1.6 mm., $E^* = 1$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.15$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 16, tr = 2.15, c = 1.6 mm., $E^* = 1$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.15$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 16, tr = 2.15, c = 1.6 mm., $E^* = 1$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.15$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 16, tr = 11.2, c = 1.6 mm., $E^* = 1$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.78$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -26 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C



Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 16, tr = 2.15, c = 1.6 mm., E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.15$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-40 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $30.50/51.00 = 0.598$

Weld Size Calculations, Description: M1-6"

Intermediate Calc. for nozzle/shell Welds	Tmin	14.3575 mm.
Intermediate Calc. for pad/shell Welds	TminPad	14.4000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.
Pad Weld	7.2000 = 0.5 * TminPad	8.4840 = 0.7 * Wp mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (16.1 - 4.28 + 2 \cdot 14.4 \cdot 0.86 \cdot (1 \cdot 14.4 - 11.2)) \cdot 138)$
 $= 17788.42 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can / 707)^2 \cdot fr2) \cdot Sv$
 $= (7.51 + 16 + 1.86 - 0 \cdot 0.86) \cdot 138$
 $= 35708.81 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (7.51 + 0 + 0.86 + (3.54)) \cdot 138$
 $= 16752.17 \text{ Kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (7.51 + 0 + 1.86 + 16 + (3.54)) \cdot 138$
 $= 40684.19 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 168 \cdot 10 \cdot 0.49 \cdot 118$
 $= 15584. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.14/2.0) \cdot 268 \cdot 12 \cdot 0.49 \cdot 138$
 $= 34867. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:



$$\begin{aligned}
 &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\
 &= (3.14 \cdot 77) \cdot (16 - 1.6) \cdot 0.7 \cdot 118 \\
 &= 29237. \text{ Kgf}
 \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\
 &= (3.14/2) \cdot 168 \cdot 16 \cdot 0.74 \cdot 138 \\
 &= 44039. \text{ Kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\
 &= (3.14/2.0) \cdot 168 \cdot (16 - 1.6) \cdot 0.74 \cdot 138 \\
 &= 39635. \text{ Kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (34867 + 29237) = 64105 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (15584 + 44039 + 39635 + 0) = 99259 \text{ Kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Spew + Tngw + Sinw) \\
 &= (34867 + 39635 + 0) = 74502 \text{ Kgf}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 64104	Kgf, must exceed W = 17788 Kgf	or W1 = 35708 Kgf
Path 2-2 = 99258	Kgf, must exceed W = 17788 Kgf	or W2 = 16752 Kgf
Path 3-3 = 74502	Kgf, must exceed W = 17788 Kgf	or W3 = 40684 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 30.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 7.1299 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 223.1299 mm.

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Input, Nozzle Desc: N6-2"

From: 20

Pressure for Reinforcement Calculations	P	30.500	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1000.00	mm.
Shell Finished (Minimum) Thickness	t	16.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		200.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: N6-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.375	in.
Actual Thickness Used in Calculation	0.301	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 30.5 \cdot 502 / (138 \cdot 1 - 0.6 \cdot 30.5)$
 $= 11.2360$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)



$$= 30.5 \cdot 30.2 / (118 \cdot 1 + 0.4 \cdot 30.5)$$
$$= 0.7717 \text{ mm.}$$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	96.4684	mm.
Parallel to Vessel Wall, opening length	d	48.2342	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	15.1135	mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: N6-2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.3717	mm.
Wall Thickness per UG16(b),	tr16b	= 3.1000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 12.8360	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 12.8360	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.1000	mm.
Wall Thickness per table UG-45	tb3	= 5.0200	mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$
$$= \min[5.02, \max(12.8, 3.1)]$$
$$= 5.0200 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$
$$= \max(2.37, 5.02)$$
$$= 5.0200 \text{ mm.}$$

Available Nozzle Neck Thickness = 7.6454 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 47.9,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 247.1 N./mm ²	Passed
Occasional	: 5.4,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 30.3,	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 7.65, tr = 0.77, c = 1.6 mm., E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.13, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 7.65, tr = 0.77, c = 1.6 mm., E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.13, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:



Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 30.50/102.00 = 0.299

Weld Size Calculations, Description: N6-2"

Intermediate Calc. for nozzle/shell Welds Tmin 6.0454 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	4.2318 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.9106 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 216.9106 mm.

Input Echo, WRC297 Item 1, Description: N6-2" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.6000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1000.000	mm.
Shell Thickness	16.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.6000	mm.
Nozzle Diameter	60.325	mm.
Nozzle Thickness	7.6454	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	30.50	bars
Radial Load	P	382.42	Kgf
Circumferential Shear	Vc	254.95	Kgf
Longitudinal Shear	Vl	254.95	Kgf
Circumferential Moment	Mc	35.68	Kg-m.
Longitudinal Moment	Ml	35.68	Kg-m.
Torsional Moment	Mt	50.97	Kg-m.
Include Axial Pressure Thrust		No	
Include Pressure Stress Indices per Div. 2		No	
Local Loads applied at end of Nozzle/Attachment		No	

Warning - The ratio, Dn/Tv, (3.770) is < 5
 check the Limitations of WRC bulletin 297.
 Warning - The ratio, Dn/Tn (7.890) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:



Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 60.325 + 2 \cdot 1.65 \cdot \sqrt{508.8(16.0 - 1.6)}$$

$$= 342.793 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1017.600 mm.

Nozzle Outside Diameter (d) = 60.325 mm.

Vessel Thickness used (T) = 14.400 mm.

Nozzle Thickness used (t) = 6.045 mm.

T / t = 2.382

d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.498$$

Nr/P = 0.199

Mr/P = 0.181

M0/P = 0.134

N0/P = 0.173

MrD/Mc = 0.212

NrDL/Mc = 0.193

M0d/Mc = 0.189

N0DL/Mc = -0.101

MrD/M1 = 0.197

NrDL/M1 = 0.186

M0D/M1 = 0.176

N0DL/M1 = -0.034

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-3	-3	-3	-3
Outplane Bending (P)	-14	14	-14	14
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-29	29	29	-29
Normal Pressure Stress	104	107	104	107
Outplane Stress Summary	58	149	115	88

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-3	-3	-3	-3
Inplane Bending (P)	-19	19	-19	19
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-33	33	33	-33
Inplane Pressure Stress	52	52	52	52
Inplane Stress Summary	-9	96	67	40

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	6	6	6	6



Shear Stress Summary 7 7 4 4

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE
 (Stress Intensities)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress	69	150	116	89
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to
 circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	-3	-3	-3	-3
Outplane Bending (P)	-14	14	-14	14
Outplane Membrane (Mc)	2	2	-2	-2
Outplane Bending (Mc)	-31	31	31	-31
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	52	52	52	52

Outplane Stress Summary 5 98 63 29

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stresses parallel to
 circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	-3	-3	-3	-3
Inplane Bending (P)	-19	19	-19	19
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-35	35	35	-35
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	104	107	104	107

Inplane Stress Summary 40 153 122 93

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Shear stress normal to
 circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	6	6	6	6

Shear Stress Summary 4 4 7 7

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stress Intensities)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	41	154	123	94
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE
 (Stresses in the
 hoop direction)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	-3	-3	-3	-3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	12	16	12	16



Hoop Stress Summary	10	14	8	12
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	-3	-3	-3	-3
Axial Bending (P)	-49	49	-49	49
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-27	-27	27	27
Axial Bending (ML)	-99	99	99	-99
Axial Pressure Stress	6	6	6	6

Axial Stress Summary	-173	124	79	-18
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	4	4	-4	-4
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	14	14	14	14

Shear Stress Summary	18	18	10	10
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	188	127	80	37

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-3	-3	-3	-3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	2	2	-2	-2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	12	16	12	16

Hoop Stress Summary	12	16	7	10
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	-3	-3	-3	-3
Axial Bending (P)	-49	49	-49	49
Axial Membrane (Mc)	-27	-27	27	27
Axial Bending (Mc)	-109	109	109	-109
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6

Axial Stress Summary	-184	134	90	-29
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0



Shear due to (V1)	-4	-4	4	4
Shear due to Torsion	14	14	14	14
Shear Stress Summary	10	10	18	18

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	197	135	94	55

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	104	107	104	107	104	107	104	107
Circ. Pl (SUS)	-3	-3	-3	-3	-8	-8	2	2
Circ. Q (SUS)	-43	43	15	-15	-54	54	16	-16
Long. Pm (SUS)	52	52	52	52	52	52	52	52
Long. Pl (SUS)	-8	-8	2	2	-1	-1	-5	-5
Long. Q (SUS)	-52	52	14	-14	-45	45	17	-17
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	104.0	107.0	104.0	107.0	104.0	107.0	104.0	107.0
Pm+Pl (SUS)	101.0	104.0	101.0	104.0	96.0	99.0	106.0	109.0
Pm+Pl+Q (Total)	67.5	147.9	116.5	89.5	42.7	153.4	122.8	93.8

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	107.00	138.00	Passed
Pm+Pl (SUS)	109.02	207.00	Passed
Pm+Pl+Q (TOTAL)	153.44	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	12	16	12	16	12	16	12	16
Circ. Pl (SUS)	-3	-3	-3	-3	-1	-1	-5	-5
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	-30	-30	24	24	-30	-30	24	24
Long. Q (SUS)	-148	148	50	-50	-158	158	60	-60
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	4	4	-4	-4	-4	-4	4	4
Shear Q (SUS)	14	14	14	14	14	14	14	14
Pm (SUS)	12.0	16.0	12.0	16.0	12.0	16.0	12.0	16.0



Pm+Pl (SUS)	34.0	37.9	30.7	30.9	35.9	39.8	30.7	30.8
Pm+Pl+Q (Total)	184.5	126.8	81.4	38.6	194.0	134.8	93.7	54.6

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	16.00	118.00	Passed
Pm+Pl (SUS)	39.81	177.00	Passed
Pm+Pl+Q (TOTAL)	194.03	354.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: K2-2"

From: 20

Pressure for Reinforcement Calculations	P	30.500	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1000.00	mm.
Shell Finished (Minimum) Thickness	t	16.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1800.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: K2-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.661	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]



$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 30.5 \cdot 502 / (138 \cdot 1 - 0.6 \cdot 30.5)$$

$$= 11.2360 \text{ mm.}$$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]

$$= R \left(\exp \left[\frac{P}{S_n \cdot E} \right] - 1 \right) \text{ per Appendix 1-2 (a)(1)}$$

$$= 27 \left(\exp \left[\frac{30.5}{(138 \cdot 1)} \right] - 1 \right)$$

$$= 0.6034 \text{ mm.}$$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.2000	mm.
Parallel to Vessel Wall	Rn+tn+t	56.6000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	36.0000	mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: K2-2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.2034	mm.
Wall Thickness per UG16(b),	tr16b	= 3.1000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 12.8360	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 12.8360	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.1000	mm.
Wall Thickness per table UG-45	tb3	= 6.4000	mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[6.4, \max(12.8, 3.1)]$$

$$= 6.4000 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(2.2, 6.4)$$

$$= 6.4000 \text{ mm.}$$

Available Nozzle Neck Thickness = 16.8000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B**

Govrn. thk, tg = 16, tr = 11.2, c = 1.6 mm., E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.78, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-26 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-40 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 30.50/51.00 = 0.598

Weld Size Calculations, Description: K2-2"

Intermediate Calc. for nozzle/shell Welds Tmin 14.4000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.



Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.7840 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 224.1590 mm.

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Input, Nozzle Desc: K1A-3"

From: 20

Pressure for Reinforcement Calculations	P	30.500	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1000.00	mm.
Shell Finished (Minimum) Thickness	t	16.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1600.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.6000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0250	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: K1A-3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.000	in.
Actual Thickness Used in Calculation	0.811	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]



$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 30.5 \cdot 502 / (138 \cdot 1 - 0.6 \cdot 30.5)$$

$$= 11.2360 \text{ mm.}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 30.5 \cdot 39.7 / (138 \cdot 1 - 0.6 \cdot 30.5)$$

$$= 0.8893 \text{ mm.}$$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	158.8000	mm.
Parallel to Vessel Wall, opening length	d	79.4000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	36.0000	mm.

Weld Strength Reduction Factor [fr1]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 138 / 138)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr2]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 138 / 138)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(1, 1)$$

$$= 1.000$$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	8.921	NA	NA
Area in Shell	A1	2.512	NA	NA
Area in Nozzle Wall	A2	13.040	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	16.552	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (79.4 \cdot 11.2 \cdot 1 + 2 \cdot 19 \cdot 11.2 \cdot 1 \cdot (1 - 1))$$

$$= 8.921 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 79.4 (1 \cdot 14.4 - 1 \cdot 11.2) - 2 \cdot 19$$

$$(1 \cdot 14.4 - 1 \cdot 11.2) \cdot (1 - 1)$$

$$= 2.512 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2$$

$$= (2 \cdot 36) (19 - 0.89) 1$$

$$= 13.040 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr2 + (W_i - can / 0.707)^2 \cdot fr2$$



$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.4893 mm.
Wall Thickness per UG16(b),	tr16b = 3.1000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 12.8360 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 12.8360 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.1000 mm.
Wall Thickness per table UG-45	tb3 = 7.3300 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[7.33, \max(12.8, 3.1)]$$

$$= 7.3300 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(2.49, 7.33)$$

$$= 7.3300 \text{ mm.}$$

Available Nozzle Neck Thickness = 20.6000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 16, tr = 11.2, c = 1.6 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.78, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-26 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-40 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 30.50/51.00 = 0.598

Weld Size Calculations, Description: K1A-3"

Intermediate Calc. for nozzle/shell Welds Tmin 14.4000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (8.92 - 2.51 + 2 \cdot 19 \cdot 1 \cdot (1 \cdot 14.4 - 11.2)) \cdot 138)$$

$$= 10710.51 \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$$

$$= (13 + 0 + 1 - 0 \cdot 1) \cdot 138$$

$$= 19756.09 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$



$$= (13 + 0 + 1 + (5.47)) \cdot 138$$

$$= 27456.05 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t_{fr1})) \cdot S$$

$$= (13 + 0 + 1 + 0 + (5.47)) \cdot 138$$

$$= 27456.05 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 117 \cdot 10 \cdot 0.49 \cdot 138$$

$$= 12715. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 49.2) \cdot (20.6 - 1.6) \cdot 0.7 \cdot 138$$

$$= 28928. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 117 \cdot (16 - 1.6) \cdot 0.74 \cdot 138$$

$$= 27652. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (12715 + 28928) = 41643 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (12715 + 0 + 27652 + 0) = 40368 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (12715 + 27652 + 0) = 40368 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 41643	Kgf, must exceed W = 10710 Kgf	or W1 = 19756 Kgf
Path 2-2 = 40367	Kgf, must exceed W = 10710 Kgf	or W2 = 27456 Kgf
Path 3-3 = 40367	Kgf, must exceed W = 10710 Kgf	or W3 = 27456 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 30.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 3.4576 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 219.4826 mm.

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Input, Nozzle Desc: K1B-3"

From: 20

Pressure for Reinforcement Calculations	P	30.500	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1000.00	mm.
Shell Finished (Minimum) Thickness	t	16.0000	mm.
Shell Internal Corrosion Allowance	c	1.6000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		250.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.6000	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0250	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: K1B-3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.000	in.
Actual Thickness Used in Calculation	0.811	in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]



$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 30.5 \cdot 502 / (138 \cdot 1 - 0.6 \cdot 30.5)$$

$$= 11.2360 \text{ mm.}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 30.5 \cdot 39.7 / (138 \cdot 1 - 0.6 \cdot 30.5)$$

$$= 0.8893 \text{ mm.}$$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	158.8000	mm.
Parallel to Vessel Wall, opening length	d	79.4000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	36.0000	mm.

Weld Strength Reduction Factor [fr1]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 138 / 138)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr2]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 138 / 138)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(1, 1)$$

$$= 1.000$$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	8.921	NA	NA
Area in Shell	A1	2.512	NA	NA
Area in Nozzle Wall	A2	13.040	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	16.552	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (79.4 \cdot 11.2 \cdot 1 + 2 \cdot 19 \cdot 11.2 \cdot 1 \cdot (1 - 1))$$

$$= 8.921 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 79.4 (1 \cdot 14.4 - 1 \cdot 11.2) - 2 \cdot 19$$

$$(1 \cdot 14.4 - 1 \cdot 11.2) \cdot (1 - 1)$$

$$= 2.512 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp) (tn - trn) fr2$$

$$= (2 \cdot 36) (19 - 0.89) 1$$

$$= 13.040 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr2 + (W_i - can / 0.707)^2 \cdot fr2$$



$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.4893 mm.
Wall Thickness per UG16(b),	tr16b = 3.1000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 12.8360 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 12.8360 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.1000 mm.
Wall Thickness per table UG-45	tb3 = 7.3300 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[7.33, \max(12.8, 3.1)]$$

$$= 7.3300 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(2.49, 7.33)$$

$$= 7.3300 \text{ mm.}$$

Available Nozzle Neck Thickness = 20.6000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 16, tr = 11.2, c = 1.6 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.78, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-26 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-40 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 30.50/51.00 = 0.598

Weld Size Calculations, Description: K1B-3"

Intermediate Calc. for nozzle/shell Welds Tmin 14.4000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (8.92 - 2.51 + 2 \cdot 19 \cdot 1 \cdot (1 \cdot 14.4 - 11.2)) \cdot 138)$$

$$= 10710.62 \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$$

$$= (13 + 0 + 1 - 0 \cdot 1) \cdot 138$$

$$= 19756.09 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$



$$= (13 + 0 + 1 + (5.47)) \cdot 138$$

$$= 27456.05 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t_{fr1})) \cdot S$$

$$= (13 + 0 + 1 + 0 + (5.47)) \cdot 138$$

$$= 27456.05 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 117 \cdot 10 \cdot 0.49 \cdot 138$$

$$= 12715. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 49.2) \cdot (20.6 - 1.6) \cdot 0.7 \cdot 138$$

$$= 28928. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 117 \cdot (16 - 1.6) \cdot 0.74 \cdot 138$$

$$= 27652. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (12715 + 28928) = 41643 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (12715 + 0 + 27652 + 0) = 40368 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (12715 + 27652 + 0) = 40368 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 41643	Kgf, must exceed W = 10710 Kgf	or W1 = 19756 Kgf
Path 2-2 = 40367	Kgf, must exceed W = 10710 Kgf	or W2 = 27456 Kgf
Path 3-3 = 40367	Kgf, must exceed W = 10710 Kgf	or W3 = 27456 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 30.5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 3.4576 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 219.4826 mm.

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Input, Nozzle Desc: N4-2"

From: 30

Pressure for Reinforcement Calculations	P	30.500	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	1000.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	16.0000	mm.
Head Internal Corrosion Allowance	c	1.6000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	100.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		120.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: N4-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.375	in.
Actual Thickness Used in Calculation	0.301	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 30.5 \cdot 1003 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 30.5)$
 $= 9.9700$ mm.



Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 30.5 \cdot 30.2 / (118 \cdot 1 + 0.4 \cdot 30.5)$
 $= 0.7717$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	96.9764	mm.
Parallel to Vessel Wall, opening length	d	48.4882	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	15.1135	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: N4-2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 7.65, tr = 0.77, c = 1.6 mm., E* = 1
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.13$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 7.65, tr = 0.77, c = 1.6 mm., E* = 1
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.13$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $30.50 / 102.00 = 0.299$

Weld Size Calculations, Description: N4-2"

Intermediate Calc. for nozzle/shell Welds	Tmin	6.0454	mm.
---	------	--------	-----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.2318 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm.

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: N4-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.375	in.
Actual Thickness Used in Calculation	0.301	in.

Nozzle input data check completed without errors.



Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
= $30.5 \cdot 1003 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 30.5)$
= 9.9700 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $30.5 \cdot 30.2 / (118 \cdot 1 + 0.4 \cdot 30.5)$
= 0.7717 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	96.4684	mm.
Parallel to Vessel Wall, opening length	d	48.2342	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	15.1135	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: N4-2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.3717	mm.
Wall Thickness per UG16(b),	tr16b	= 3.1000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 12.6645	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 12.6645	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.1000	mm.
Wall Thickness per table UG-45	tb3	= 5.0200	mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[5.02, max(12.7, 3.1)]
= 5.0200 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(2.37, 5.02)
= 5.0200 mm.

Available Nozzle Neck Thickness = 7.6454 mm. -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code,
(see 319.4.4 and 302.3.5):**

Sustained	: 47.9,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 247.1 N./mm ²	Passed
Occasional	: 5.4,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 30.3,	Allowable	: 82.6 N./mm ²	Passed

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of
the expansion stress allowable.*

Weld Size Calculations, Description: N4-2"

Intermediate Calc. for nozzle/shell Welds Tmin 6.0454 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	4.2318 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm.

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 3.8340 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 169.9291 mm.

Warning:
The equations for nozzle local stress analysis pertain to radial nozzles only



and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC107/537 Item 1, Description: N4-2" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.6000	mm.
Vessel Diameter	Dv	1800.000	mm.
Vessel Thickness	Tv	16.000	mm.
Design Temperature	T1	82.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	1.6000	mm.
Nozzle Diameter	Dn	60.325	mm.
Nozzle Thickness	Tn	7.645	mm.
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm ²
Design Internal Pressure	Dp	30.500	bars
Include Pressure Thrust		No	

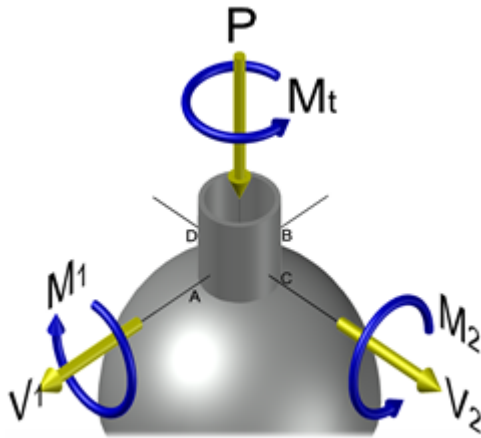
External Forces and Moments in WRC 107/537 Convention:

Radial Load	(SUS)	P	382.4	Kgf
Longitudinal Shear	(SUS)	(V1) V1	254.9	Kgf
Circumferential Shear	(SUS)	(Vc) V2	254.9	Kgf
Circumferential Moment	(SUS)	(Mc) M1	35.7	Kg-m.
Longitudinal Moment	(SUS)	(Ml) M2	35.7	Kg-m.
Torsional Moment	(SUS)	Mt	51.0	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 60.325 + 2 \cdot 1.65 \cdot \sqrt{(908.8 (16.0 - 1.6))}$$

$$= 437.836 \text{ mm.}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	382.4	Kgf
Circumferential Shear	(VC) V2	254.9	Kgf
Longitudinal Shear	(VL) V1	254.9	Kgf
Circumferential Moment	(MC) M1	35.7	Kg-m.
Longitudinal Moment	(ML) M2	35.7	Kg-m.
Torsional Moment	MT	51.0	Kg-m.

Dimensionless Param: U = 0.26 TAU = 5.00 (4.49) RHO = 2.38

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 3	0.11165	(A,B,C,D)
$M(x) / P$	SP 3	0.07600	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 3	0.32076	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 3	0.24281	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 3	0.32076	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 3	0.24281	(A,B,C,D)
$N(y) * T / P$	SP 3	0.25357	(A,B,C,D)
$M(y) / P$	SP 3	0.10127	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 3	0.10100	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 3	0.46546	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 3	0.10100	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 3	0.46546	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	-2.0	-2.0	-2.0	-2.0	-2.0	-2.0	-2.0	-2.0
Rad.	Bend. P	-8.2	8.2	-8.2	8.2	-8.2	8.2	-8.2	8.2
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-4.7	-4.7	4.7	4.7
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-21.5	21.5	21.5	-21.5
Rad.	Memb. ML	-4.7	-4.7	4.7	4.7	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-21.5	21.5	21.5	-21.5	0.0	0.0	0.0	0.0



Tot. Rad. Str.	-36.5	23.0	16.0	-10.5	-36.5	23.0	16.0	-10.5
Tang. Memb. P	-4.6	-4.6	-4.6	-4.6	-4.6	-4.6	-4.6	-4.6
Tang. Bend. P	-11.0	11.0	-11.0	11.0	-11.0	11.0	-11.0	11.0
Tang. Memb. MC	0.0	0.0	0.0	0.0	-1.5	-1.5	1.5	1.5
Tang. Bend. MC	0.0	0.0	0.0	0.0	-41.2	41.2	41.2	-41.2
Tang. Memb. ML	-1.5	-1.5	1.5	1.5	0.0	0.0	0.0	0.0
Tang. Bend. ML	-41.2	41.2	41.2	-41.2	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-58.3	46.1	27.1	-33.3	-58.3	46.1	27.1	-33.3
Shear VC	1.8	1.8	-1.8	-1.8	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-1.8	-1.8	1.8	1.8
Shear MT	6.1	6.1	6.1	6.1	6.1	6.1	6.1	6.1
Tot. Shear	7.9	7.9	4.2	4.2	4.2	4.2	7.9	7.9
Str. Int.	60.8	48.6	28.5	34.1	59.1	46.9	31.2	35.8

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		95.5	95.5	95.5	95.5	95.5	95.5	95.5	95.5
Rad. Pl (SUS)		-6.8	-6.8	2.7	2.7	-6.8	-6.8	2.7	2.7
Rad. Q (SUS)		-29.7	29.7	13.2	-13.2	-29.7	29.7	13.2	-13.2
Long. Pm (SUS)		95.5	95.5	95.5	95.5	95.5	95.5	95.5	95.5
Long. Pl (SUS)		-6.1	-6.1	-3.1	-3.1	-6.1	-6.1	-3.1	-3.1
Long. Q (SUS)		-52.2	52.2	30.2	-30.2	-52.2	52.2	30.2	-30.2
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		1.8	1.8	-1.8	-1.8	-1.8	-1.8	1.8	1.8
Shear Q (SUS)		6.1	6.1	6.1	6.1	6.1	6.1	6.1	6.1
Pm (SUS)		95.5	95.5	95.5	95.5	95.5	95.5	95.5	95.5
Pm+Pl (SUS)		91.0	91.0	98.7	98.7	91.0	91.0	98.7	98.7
Pm+Pl+Q (Total)		61.6	144.1	124.0	85.7	59.8	142.4	126.7	87.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	95.50	138.00	Passed
Pm+Pl (SUS)	98.75	207.00	Passed
Pm+Pl+Q (TOTAL)	144.05	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N3-2"

From: 30

Pressure for Reinforcement Calculations	P	30.500	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	1000.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	16.0000	mm.
Head Internal Corrosion Allowance	c	1.6000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	150.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: N3-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.375	in.
Actual Thickness Used in Calculation	0.301	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 30.5 \cdot 1003 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 30.5)$
 $= 9.9700$ mm.



Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $30.5 \cdot 30.2 / (118 \cdot 1 + 0.4 \cdot 30.5)$
= 0.7717 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	97.6592	mm.
Parallel to Vessel Wall, opening length	d	48.8296	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	15.1135	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: N3-2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 7.65, tr = 0.77, c = 1.6 mm., E* = 1
Thickness Ratio = $tr \cdot E* / (tg - c) = 0.13$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 7.65, tr = 0.77, c = 1.6 mm., E* = 1
Thickness Ratio = $tr \cdot E* / (tg - c) = 0.13$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $30.50 / 102.00 = 0.299$

Weld Size Calculations, Description: N3-2"

Intermediate Calc. for nozzle/shell Welds	Tmin	6.0454	mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.2318 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm.

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: N3-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.375	in.
Actual Thickness Used in Calculation	0.301	in.

Nozzle input data check completed without errors.



Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
= $30.5 \cdot 1003 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 30.5)$
= 9.9700 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $30.5 \cdot 30.2 / (118 \cdot 1 + 0.4 \cdot 30.5)$
= 0.7717 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	96.4684	mm.
Parallel to Vessel Wall, opening length	d	48.2342	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	15.1135	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: N3-2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.3717	mm.
Wall Thickness per UG16(b),	tr16b	= 3.1000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 12.6645	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 12.6645	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.1000	mm.
Wall Thickness per table UG-45	tb3	= 5.0200	mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[5.02, max(12.7, 3.1)]
= 5.0200 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(2.37, 5.02)
= 5.0200 mm.

Available Nozzle Neck Thickness = 7.6454 mm. -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code,
(see 319.4.4 and 302.3.5):**

Sustained	: 47.9,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 247.1 N./mm ²	Passed
Occasional	: 5.4,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 30.3,	Allowable	: 82.6 N./mm ²	Passed

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of
the expansion stress allowable.*

Weld Size Calculations, Description: N3-2"

Intermediate Calc. for nozzle/shell Welds Tmin 6.0454 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	4.2318 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm.

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 5.5471 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 171.7638 mm.

Warning:
The equations for nozzle local stress analysis pertain to radial nozzles only



and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC107/537 Item 1, Description: N3-2 :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.6000	mm.
Vessel Diameter	Dv	1800.000	mm.
Vessel Thickness	Tv	16.000	mm.
Design Temperature	T1	82.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	1.6000	mm.
Nozzle Diameter	Dn	60.325	mm.
Nozzle Thickness	Tn	7.645	mm.
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm ²
Design Internal Pressure	Dp	30.500	bars
Include Pressure Thrust		No	

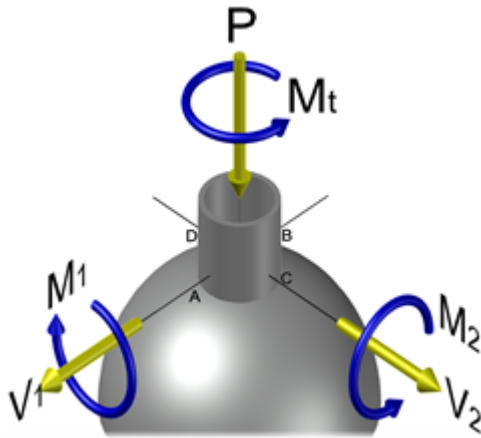
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	382.4	Kgf
Longitudinal Shear (SUS)	(V1) V1	254.9	Kgf
Circumferential Shear (SUS)	(Vc) V2	254.9	Kgf
Circumferential Moment (SUS)	(Mc) M1	35.7	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	35.7	Kg-m.
Torsional Moment (SUS)	Mt	51.0	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 60.325 + 2 \cdot 1.65 \cdot \sqrt{(908.8 (16.0 - 1.6))}$$

$$= 437.836 \text{ mm.}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	382.4	Kgf
Circumferential Shear	(VC) V2	254.9	Kgf
Longitudinal Shear	(VL) V1	254.9	Kgf
Circumferential Moment	(MC) M1	35.7	Kg-m.
Longitudinal Moment	(ML) M2	35.7	Kg-m.
Torsional Moment	MT	51.0	Kg-m.

Dimensionless Param: U = 0.26 TAU = 5.00 (4.49) RHO = 2.38

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 3	0.11165	(A,B,C,D)
$M(x) / P$	SP 3	0.07600	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 3	0.32076	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 3	0.24281	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 3	0.32076	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 3	0.24281	(A,B,C,D)
$N(y) * T / P$	SP 3	0.25357	(A,B,C,D)
$M(y) / P$	SP 3	0.10127	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 3	0.10100	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 3	0.46546	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 3	0.10100	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 3	0.46546	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	-2.0	-2.0	-2.0	-2.0	-2.0	-2.0	-2.0	-2.0
Rad.	Bend. P	-8.2	8.2	-8.2	8.2	-8.2	8.2	-8.2	8.2
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-4.7	-4.7	4.7	4.7
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-21.5	21.5	21.5	-21.5
Rad.	Memb. ML	-4.7	-4.7	4.7	4.7	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-21.5	21.5	21.5	-21.5	0.0	0.0	0.0	0.0



Tot. Rad. Str.	-36.5	23.0	16.0	-10.5	-36.5	23.0	16.0	-10.5
Tang. Memb. P	-4.6	-4.6	-4.6	-4.6	-4.6	-4.6	-4.6	-4.6
Tang. Bend. P	-11.0	11.0	-11.0	11.0	-11.0	11.0	-11.0	11.0
Tang. Memb. MC	0.0	0.0	0.0	0.0	-1.5	-1.5	1.5	1.5
Tang. Bend. MC	0.0	0.0	0.0	0.0	-41.2	41.2	41.2	-41.2
Tang. Memb. ML	-1.5	-1.5	1.5	1.5	0.0	0.0	0.0	0.0
Tang. Bend. ML	-41.2	41.2	41.2	-41.2	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-58.3	46.1	27.1	-33.3	-58.3	46.1	27.1	-33.3
Shear VC	1.8	1.8	-1.8	-1.8	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-1.8	-1.8	1.8	1.8
Shear MT	6.1	6.1	6.1	6.1	6.1	6.1	6.1	6.1
Tot. Shear	7.9	7.9	4.2	4.2	4.2	4.2	7.9	7.9
Str. Int.	60.8	48.6	28.5	34.1	59.1	46.9	31.2	35.8

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		95.5	95.5	95.5	95.5	95.5	95.5	95.5	95.5
Rad. Pl (SUS)		-6.8	-6.8	2.7	2.7	-6.8	-6.8	2.7	2.7
Rad. Q (SUS)		-29.7	29.7	13.2	-13.2	-29.7	29.7	13.2	-13.2
Long. Pm (SUS)		95.5	95.5	95.5	95.5	95.5	95.5	95.5	95.5
Long. Pl (SUS)		-6.1	-6.1	-3.1	-3.1	-6.1	-6.1	-3.1	-3.1
Long. Q (SUS)		-52.2	52.2	30.2	-30.2	-52.2	52.2	30.2	-30.2
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		1.8	1.8	-1.8	-1.8	-1.8	-1.8	1.8	1.8
Shear Q (SUS)		6.1	6.1	6.1	6.1	6.1	6.1	6.1	6.1
Pm (SUS)		95.5	95.5	95.5	95.5	95.5	95.5	95.5	95.5
Pm+Pl (SUS)		91.0	91.0	98.7	98.7	91.0	91.0	98.7	98.7
Pm+Pl+Q (Total)		61.6	144.1	124.0	85.7	59.8	142.4	126.7	87.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	95.50	138.00	Passed
Pm+Pl (SUS)	98.75	207.00	Passed
Pm+Pl+Q (TOTAL)	144.05	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N5-2"

From: 30

Pressure for Reinforcement Calculations	P	30.500	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Elliptical Head	D	1000.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	16.0000	mm.
Head Internal Corrosion Allowance	c	1.6000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	150.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		240.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.6000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: N5-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.375	in.
Actual Thickness Used in Calculation	0.301	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 30.5 \cdot 1003 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 30.5)$
 $= 9.9700$ mm.



Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $30.5 \cdot 30.2 / (118 \cdot 1 + 0.4 \cdot 30.5)$
= 0.7717 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	97.6592	mm.
Parallel to Vessel Wall, opening length	d	48.8296	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	15.1135	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: N5-2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 7.65, tr = 0.77, c = 1.6 mm., E* = 1
Thickness Ratio = $tr \cdot E* / (tg - c) = 0.13$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 7.65, tr = 0.77, c = 1.6 mm., E* = 1
Thickness Ratio = $tr \cdot E* / (tg - c) = 0.13$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-40 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $30.50 / 51.00 = 0.598$

Weld Size Calculations, Description: N5-2"

Intermediate Calc. for nozzle/shell Welds	Tmin	6.0454	mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.2318 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm.

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: N5-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	2.375	in.
Actual Thickness Used in Calculation	0.301	in.

Nozzle input data check completed without errors.



Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 30.5 \cdot 1003 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 30.5)$
 $= 9.9700$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 30.5 \cdot 30.2 / (118 \cdot 1 + 0.4 \cdot 30.5)$
 $= 0.7717$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	96.4684	mm.
Parallel to Vessel Wall, opening length	d	48.2342	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	15.1135	mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: N5-2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.3717	mm.
Wall Thickness per UG16(b),	tr16b	= 3.1000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 12.6645	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 12.6645	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.1000	mm.
Wall Thickness per table UG-45	tb3	= 5.0200	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.02, \max(12.7, 3.1)]$
 $= 5.0200$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.37, 5.02)$
 $= 5.0200$ mm.

Available Nozzle Neck Thickness = 7.6454 mm. -> OK

Weld Size Calculations, Description: N5-2"

Intermediate Calc. for nozzle/shell Welds	Tmin	6.0454	mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.2318 = 0.7 \cdot tmin.$	$7.0700 = 0.7 \cdot Wo$ mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 5.5471 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 171.7638 mm.

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